

1 Dedekind domains

Definition 1.1. Let A be a ring. We say that A is a discrete valuation ring (DVR) if A is a principal ideal domain with a unique non-zero prime ideal $\mathfrak{m}_A \subset A$.

If A is a DVR, then $k_A = A/\mathfrak{m}_A$ is a field, its residue field. If $\pi \in \mathfrak{m}_A$ is a generator, we call π a uniformizer. If π' is another uniformizer then $\pi' = u\pi$ for some $u \in A^\times$.

If A is a DVR with $\text{Frac } A = K$, then any element $x = a/b \in K^\times$ admits a unique expression as $x = \pi^n u$, with $n \in \mathbb{Z}$ and $u \in A^\times$. We call $v(x) = n$ the valuation of the element x . The function $v : K^\times \rightarrow \mathbb{Z}$ satisfies the following conditions:

1. $v : K^\times \rightarrow \mathbb{Z}$ is a surjective homomorphism.
2. If $x, y \in K^\times$ and $x + y \in K^\times$ then we have $v(x + y) \geq \min(v(x), v(y))$.

If K is any field and $v : K^\times \rightarrow \mathbb{Z}$ is a function satisfying these conditions, then v is called a valuation. We can then define $A = \{x \in K \mid v(x) \geq 0\}$; then A is a DVR with non-zero prime ideal $\mathfrak{m}_A = \{x \in K \mid v(x) > 0\}$.

Lemma 1.2. Let K be a field. Then this process defines a bijection between the set of subrings $A \subset K$ which are discrete valuation rings and the set of valuations $v : K^\times \rightarrow \mathbb{Z}$.

Definition 1.3. A ring A is called a local ring if it has a unique maximal ideal.

Note that a DVR is a local ring.

Lemma 1.4 (Nakayama's lemma). Let A be a local ring with maximal ideal $\mathfrak{m}$, and let M be a finitely generated A -module.

1. If $\mathfrak{m}M = M$, then $M = 0$.
2. If $N \subset M$ is a submodule such that $N \rightarrow M/\mathfrak{m}M$ is surjective, then $N = M$.

Proof. We observe that $A - \mathfrak{m} = A^\times$. For the first part, suppose that M is generated by elements $m_1, \dots, m_n$, and that $n \geq 1$ is minimal with respect to this property. We can write $m_n = a_1 m_1 + \dots + a_n m_n$ with $a_i \in \mathfrak{m}$, hence $(1 - a_n)m_n = a_1 m_1 + \dots + a_{n-1} m_{n-1}$. Since $1 - a_n$ is a unit, we conclude that A is generated by $m_1, \dots, m_{n-1}$, a contradiction to the minimality of n .

For the second part, the hypothesis implies $M = N + \mathfrak{m}M$. We apply the first part to the quotient M/N to conclude $M/N = 0$, hence $M = N$. $\square$

Proposition 1.5. Let A be a Noetherian domain. Then the following are equivalent:

1. A is a DVR.
2. A is integrally closed and has exactly one non-zero prime ideal.

We recall that to say that A is integrally closed is to say that for any element $\gamma \in K = \text{Frac } A$ satisfying a relation $\gamma^n + a_1\gamma^{n-1} + \cdots + a_n = 0$ with $a_1, \dots, a_n \in A$ (and $n \geq 1$), we have $\gamma \in A$.

If $\gamma \in K$, then γ satisfies such a relation if and only if $A[\gamma]$ is a finitely generated A -module. Indeed, if γ satisfies a relation of this type then $1, \gamma, \dots, \gamma^{n-1}$ generates $A[\gamma]$ as A -module. Conversely, suppose that $A[\gamma]$ is finitely generated. Using that A is Noetherian, we see that the sequence $A, A + A\gamma, A + A\gamma + A\gamma^2, \dots$ of A -submodules of $A[\gamma]$ must eventually be stationary. If $\gamma^n \in A + A\gamma + \cdots + A\gamma^{n-1}$, then γ is a zero of a monic polynomial with coefficients in A .

Proof. If A is a DVR, then A is integrally closed: if an element $a = u\pi^r$ satisfies a relation $a^n + a_1a^{n-1} + \cdots + a_n = 0$, then after multiplying through by u^{-n} , we can assume $a = \pi^r$. If $r < 0$ then we get $1 = a_1\pi^{-r} + \cdots + a_n\pi^{-nr}$. The left-hand side lies in $\mathfrak{m}_A$, a contradiction. It clearly has a unique non-zero prime ideal.

Suppose conversely that A is a Noetherian domain which is integrally closed, and has exactly one non-zero prime ideal $\mathfrak{p}$. Note that if $\mathfrak{a} \subset A$ is a proper non-zero ideal, then we can find $n \geq 1$ such that $\mathfrak{p}^n \subset \mathfrak{a} \subset \mathfrak{p}$. Indeed, if not then we can find a proper ideal $\mathfrak{a} \subset A$ maximal with respect to the property that it contains no $\mathfrak{p}^n$. Then $\mathfrak{a} \neq \mathfrak{p}$, so we can find $a, b \in A$ such that $ab \in \mathfrak{a}$ but $a \notin \mathfrak{a}$ and $b \notin \mathfrak{a}$. The inclusions $\mathfrak{a} \subset \mathfrak{a} + (a)$ and $\mathfrak{a} \subset \mathfrak{a} + (b)$ are then proper, so we can find $m \geq 1$ such that $\mathfrak{p}^m \subset \mathfrak{a} + (a)$ and $\mathfrak{p}^m \subset \mathfrak{a} + (b)$, hence $\mathfrak{p}^{2m} \subset (\mathfrak{a} + (a))(\mathfrak{a} + (b)) \subset \mathfrak{a}$, a contradiction.

Let us therefore choose a non-zero element $y \in \mathfrak{p}$, and $n \geq 1$ such that $\mathfrak{p}^n \subset (y) \subset \mathfrak{p}$. We can choose $n \geq 1$ minimal with respect to this property. We are going to show that $\mathfrak{p}$ is principal. If $n = 1$, then we're done. Otherwise $n \geq 2$, and we have $\mathfrak{p}^{n-1} \not\subset (y)$, and we can choose $x \in \mathfrak{p}^{n-1} - (y)$. Let $z = x/y \in K$. We have $x\mathfrak{p} \subset \mathfrak{p}^n \subset (y)$, hence $z\mathfrak{p} \subset A$ is an ideal. If $z\mathfrak{p} \subset \mathfrak{p}$, then $A[z]$ injects into $\text{End}_A(\mathfrak{p})$, a finitely generated A -module (since A is Noetherian). Using that A is integrally closed, we obtain $z \in A$, hence $x = yz \in (y)$, a contradiction. Thus $z\mathfrak{p} = A$, and $\mathfrak{p} = z^{-1}A$ and $\mathfrak{p}$ is principal.

Let $\pi = z^{-1}$. We claim that every element $a \in A$ admits a unique expression $a = u\pi^n$ with $u \in A^\times$, which will show that A is a PID. The uniqueness is clear. To show existence, it is enough to show that $\mathfrak{q} = \bigcap_{n \geq 1} (\pi^n A) = 0$ (as then $A = \bigcup_{n \geq 0} (\pi^n A - \pi^{n+1} A)$). However, we have $\pi\mathfrak{q} = \mathfrak{q}$, hence $\mathfrak{q} = 0$ (by Nakayama's lemma). This completes the proof. $\square$

We are going to study rings which are 'locally DVRs'. We first need to define localization.

Definition 1.6. Let A be a ring. A multiplicative subset $S \subset A$ is by definition a subset containing 1 and which is closed under multiplication.

We define $S^{-1}A$ to be the set of equivalence classes of pairs (a, s) with $a \in A$, $s \in S$, with respect to the equivalence relation $(a, s) \sim (b, t)$ if there exists $u \in S$ such that $u(at - bs) = 0$. We represent the equivalence class of (a, s) using the symbol a/s .

Lemma 1.7. 1. This is an equivalence relation, and $S^{-1}A$ becomes a ring with the operations $a/s + b/t = (at + sb)/(st)$ and $a/s \cdot b/t = (ab)/(st)$.

2. The map $A \rightarrow S^{-1}A$, $a \mapsto a/1$ is a ring homomorphism with kernel $\{a \in A \mid \exists s \in S, sa = 0\}$.
3. If A is an integral domain, then $\text{Frac } A = (A - \{0\})^{-1}A$. If further $S \subset A - \{0\}$, then $S^{-1}A$ may be naturally identified with a subring of $\text{Frac } A$.

Proof. The symmetry and reflexivity of the relation is clear. To show transitivity, suppose given (a, s) , (a', s') and (a'', s'') , together with $u, u' \in S$ such that $u(as' - a's) = 0$ and $u'(a's'' - a''s') = 0$. We then have $uu's'sas'' = u's''ua's = usu'a''s'$, showing that $a/s = a''/s''$ in $S^{-1}A$. The ring axioms are easily verified.

It is clear that the map $A \rightarrow S^{-1}A$ is a ring homomorphism. We have $a/1 = 0/1$ if and only if there exists $s \in S$ such that $as = 0$, as in the statement of the lemma.

If A is an integral domain, then $a/s = a'/s'$ if and only if $as' = a's$ (because A has no zero divisors). This shows that $(A - \{0\})^{-1}A = \text{Frac } A$, by definition, and that the map $S^{-1}A \rightarrow \text{Frac } A$, $a/s \mapsto a/s$, is a well-defined injection. $\square$

A common choice of S is $S = \{1, f, f^2, \dots\}$; then we write $S^{-1}A = A[1/f]$. Another common choice is $S = A - \mathfrak{p}$, where $\mathfrak{p}$ is a prime ideal of A . In fact, a proper ideal $I \subset A$ is prime if and only if $A - I$ is a multiplicative subset, as follows from the definitions.

Definition 1.8. Let A be a ring with multiplicative subset S , and let M be an A -module. We define the localized module $S^{-1}M$ to be the set of equivalence classes of pairs (m, s) with $m \in M$ and $s \in S$. Two pairs (m, s) and (m', s') are said to be equivalent if there exists $u \in S$ such that $u(s'm - sm') = 0$. The equivalence class of the pair (m, s) is denoted using the symbol m/s .

Again, one shows that $S^{-1}M$ is an $S^{-1}A$ -module, with multiplication $a/s \cdot m/t = (am)/(st)$. Localization is a functor: if M, N are A -modules and $f : M \rightarrow N$ is a homomorphism of A -modules, then there is a natural map $S^{-1}f : S^{-1}M \rightarrow S^{-1}N$ given by $S^{-1}f(m/s) = f(m)/s$. Similarly, if there is a ring homomorphism $g : A \rightarrow B$, then $S^{-1}B$ is naturally a ring and the map $S^{-1}g : S^{-1}A \rightarrow S^{-1}B$ is a ring homomorphism. An important feature of localization is its exactness:

Lemma 1.9. Let $M' \xrightarrow{f} M \xrightarrow{g} M''$ be an exact sequence of A -modules (i.e. $\ker g = \text{im } f$). Then the sequence $S^{-1}M' \xrightarrow{S^{-1}f} S^{-1}M \xrightarrow{S^{-1}g} S^{-1}M''$ is exact.

Proof. Since localization is functorial, we have $S^{-1}g \circ S^{-1}f = S^{-1}(gf) = 0$. This shows that $\text{im } S^{-1}f \subset \ker S^{-1}g$. For the other inclusion, suppose $S^{-1}g(m/s) = g(m)/s = 0$. Then there exists $u \in S$ such that $ug(m) = g(um) = 0$, hence $m' \in M'$ such that $f(m') = um$, hence $S^{-1}f(m'/us) = um/us = m/s$. $\square$

This shows, for example, that if $\mathfrak{a} \subset A$ is an ideal, then $S^{-1}\mathfrak{a}$ can be naturally identified with an ideal of $S^{-1}A$ (which is the ideal $\mathfrak{a}S^{-1}A$ generated by $\mathfrak{a}$).

Lemma 1.10. Let A be a ring, and let $f : A \rightarrow S^{-1}A$ be the natural map. The maps $\mathfrak{p} \mapsto \mathfrak{p} \cdot S^{-1}A$, $\mathfrak{q} \mapsto f^{-1}(\mathfrak{q})$ define inclusion-preserving bijections between the set of prime ideals of A which do not intersect S and the set of prime ideals of $S^{-1}A$.

Proof. We first check that $\mathfrak{p} \cdot S^{-1}A$ is indeed a prime ideal. The map $A \rightarrow A/\mathfrak{p}$ gives a map $S^{-1}A \rightarrow S^{-1}(A/\mathfrak{p})$, and the latter is non-zero since $A/\mathfrak{p}$ is non-zero and no element is killed by an element of S . An element a/s is in the kernel if and only if there exists $u \in S$ such that $ua = 0$ in $A/\mathfrak{p}$, if and only if $a \in \mathfrak{p}$. Thus $\mathfrak{p} \cdot S^{-1}A$ is the kernel of this map, and is a prime ideal of $S^{-1}A$.

We now show that these maps are mutually inverse. If $\mathfrak{p} \subset S^{-1}A$ is a prime ideal, then $f f^{-1}(\mathfrak{p}) \subset \mathfrak{p}$, hence $f^{-1}(\mathfrak{p}) \cdot S^{-1}A \subset \mathfrak{p}$. On the other hand, if $a/s \in \mathfrak{p}$ then $a/1 \in \mathfrak{p}$, hence $a \in f^{-1}(\mathfrak{p})$ and $a/s \in f^{-1}(\mathfrak{p}) \cdot S^{-1}A$. This shows one direction. In the other direction, suppose that $\mathfrak{q} \subset A$ is a prime ideal disjoint from S . Then $\mathfrak{q} \subset f^{-1}(\mathfrak{q} \cdot S^{-1}A)$. On the other hand, if $a \in A$ and $a/1 \in \mathfrak{q} \cdot S^{-1}A$, then $a/1 = qb/s$ for some $q \in \mathfrak{q}$, $b \in A$, $s \in S$, and hence there exists $u \in S$ such that $u(as - qb) = 0$, hence $uas = qbu \in \mathfrak{q}$. Since $\mathfrak{q}$ is prime and disjoint from S , we conclude $a \in \mathfrak{q}$, showing that $f^{-1}(\mathfrak{q} \cdot S^{-1}A) \subset \mathfrak{q}$, as required. $\square$

This shows that if $\mathfrak{p} \subset A$ is a prime ideal and $S = A - \mathfrak{p}$, then $S^{-1}A$ has a unique maximal ideal, namely $S^{-1}\mathfrak{p}$, and is therefore a local ring. We will write $A_{\mathfrak{p}} = (A - \mathfrak{p})^{-1}A$ for this localization; its residue field is naturally identified with $\text{Frac}(A/\mathfrak{p})$.

Proposition 1.11. *Let A be a Noetherian domain with field of fractions K . Then the following are equivalent:*

1. *For all non-zero prime ideals $\mathfrak{p} \subset A$, $A_{\mathfrak{p}}$ is a discrete valuation ring.*
2. *A is integrally closed and every non-zero prime ideal $\mathfrak{p} \subset A$ is a maximal ideal.*

A ring A satisfying these equivalent conditions is called a Dedekind domain.

Proof. We may suppose that A has non-zero prime ideals (equivalently, that A is not a field), as both conditions hold vacuously otherwise.

First suppose that A satisfies the first condition. If $\mathfrak{q} \subset \mathfrak{p}$ are prime ideals of A , then let $S = A - \mathfrak{p}$. We have $S^{-1}\mathfrak{q} \subset S^{-1}\mathfrak{p}$, hence either $\mathfrak{q} = 0$ or $\mathfrak{q} = \mathfrak{p}$. This shows that if $\mathfrak{p} \neq 0$ then $\mathfrak{p}$ is maximal. If $x \in K$ is integral over A , then it lies in $A_{\mathfrak{p}}$ for every non-zero prime $\mathfrak{p} \subset A$, hence we can write $x = a_{\mathfrak{p}}/s_{\mathfrak{p}}$, where $a_{\mathfrak{p}} \in A$ and $s_{\mathfrak{p}} \in A - \mathfrak{p}$ depend on $\mathfrak{p}$. The ideal $\mathfrak{a}$ generated by all $s_{\mathfrak{p}}$ is the unit ideal, because it is contained in no non-zero prime ideal of A ; hence we can write $1 = \sum_{\mathfrak{p}} s_{\mathfrak{p}} t_{\mathfrak{p}}$, where only finitely many $t_{\mathfrak{p}}$ are non-zero. We then have $x = \sum x s_{\mathfrak{p}} t_{\mathfrak{p}} = \sum a_{\mathfrak{p}} t_{\mathfrak{p}} \in A$, showing that A is integrally closed in K .

Now suppose instead that A satisfies the second condition. If $\mathfrak{p} \subset A$ is a non-zero prime ideal, then the ring $A_{\mathfrak{p}}$ is a local ring with a unique non-zero prime ideal, and we must show (by the previous proposition) that it is integrally closed. If $\gamma \in \text{Frac } A_{\mathfrak{p}} = K$ satisfies an equation $\gamma^n + a_1/s_1 \cdot \gamma^{n-1} + \cdots + a_n/s_n = 0$, where $a_1, \dots, a_n \in A$ and $s_1, \dots, s_n \in A - \mathfrak{p}$, then the element $\gamma(s_1 \dots s_n)$ is integral over A (by clearing denominators), so lies in A . Then $\gamma = (\gamma s_1 \dots s_n)/(s_1 \dots s_n)$ lies in $A_{\mathfrak{p}}$, as required. $\square$

Definition 1.12. *Let A be an integral domain, and let $K = \text{Frac } A$. A fractional ideal of A is by definition a finitely generated A -submodule of K .*

Lemma 1.13. *Let A be a Noetherian integral domain with field of fractions K and let $S \subset A - \{0\}$ be a multiplicative subset. Let $\mathfrak{a}, \mathfrak{b} \subset K$ be fractional ideals. Then:*

$$1. (S^{-1}\mathfrak{a})(S^{-1}\mathfrak{b}) = S^{-1}(\mathfrak{a}\mathfrak{b}) \text{ and } S^{-1}\mathfrak{a} + S^{-1}\mathfrak{b} = S^{-1}(\mathfrak{a} + \mathfrak{b}).$$

2. Define $(\mathfrak{a} : \mathfrak{b}) = \{x \in K \mid x\mathfrak{b} \subset \mathfrak{a}\}$, and suppose that $\mathfrak{b}$ is non-zero. Then $(\mathfrak{a} : \mathfrak{b})$ is a fractional ideal of A and $S^{-1}(\mathfrak{a} : \mathfrak{b}) = (S^{-1}\mathfrak{a} : S^{-1}\mathfrak{b})$.

Proof. Suppose that $\mathfrak{b} = (b_1, \dots, b_n)$, where the b_i are non-zero. Then $(\mathfrak{a} : \mathfrak{b}) = \cap_{i=1}^n b_i^{-1}\mathfrak{a}$, which shows that $(\mathfrak{a} : \mathfrak{b})$ is finitely generated (as A is Noetherian), hence a fractional ideal of A . The other assertions follow from the fact that localization commutes with finite sums and intersections of submodules. $\square$

Proposition 1.14. *Let A be a Dedekind domain. Then the set $\text{Div } A$ of non-zero fractional ideals of A forms a group under multiplication.*

Proof. Multiplication of fractional ideals gives an associative composition law in $\text{Div } A$, with identity $(1) = A$. It remains to show the existence of inverses. We claim that for any non-zero fractional ideal $\mathfrak{a} \subset A$, we have $\mathfrak{a} \cdot (A : \mathfrak{a}) = A$. Since $(A : \mathfrak{a})$ is a fractional ideal, this will be enough. If A is a DVR then we have $\mathfrak{a} = \pi^n A$ for some $n \in \mathbb{Z}$, and $(A : \mathfrak{a}) = \pi^{-n} A$; so this is clear.

In general, we have for any non-zero prime $\mathfrak{p} \subset A$,

$$(\mathfrak{a}(A : \mathfrak{a}))_{\mathfrak{p}} = \mathfrak{a}_{\mathfrak{p}}(A : \mathfrak{a})_{\mathfrak{p}} = A_{\mathfrak{p}},$$

by the lemma. It therefore suffices to show that for any fractional ideal $\mathfrak{b} \subset K$, if $\mathfrak{b}_{\mathfrak{p}} = A_{\mathfrak{p}}$ for all non-zero prime ideals $\mathfrak{p}$ of A , then $\mathfrak{b} = A$. Let us show the more general statement that if $\mathfrak{b}, \mathfrak{c}$ are fractional ideals such that $\mathfrak{b}_{\mathfrak{p}} \subset \mathfrak{c}_{\mathfrak{p}}$ for all $\mathfrak{p}$, then $\mathfrak{b} \subset \mathfrak{c}$.

Let $b \in \mathfrak{b}$. For all non-zero prime ideals $\mathfrak{p} \subset A$, we can find an expression $b = c_{\mathfrak{p}}/s_{\mathfrak{p}}$ with $c_{\mathfrak{p}} \in \mathfrak{c}$ and $s_{\mathfrak{p}} \in A - \mathfrak{p}$. The ideal $(s_{\mathfrak{p}})$ is the unit ideal, since it can be contained in no maximal ideal of A ; we can therefore find non-zero prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_n$ and $a_i \in A$ such that $1 = \sum_{i=1}^n a_i s_{\mathfrak{p}_i}$. Multiplying through, we obtain

$$b = \sum_{i=1}^n a_i b s_{\mathfrak{p}_i} = \sum_{i=1}^n a_i c_{\mathfrak{p}_i} \in \mathfrak{c},$$

as desired. $\square$

Lemma 1.15. *Let A be a Dedekind domain. Then for each non-zero ideal $\mathfrak{a} \subset A$, there are only finitely many non-zero prime ideals $\mathfrak{p} \subset A$ such that $\mathfrak{a} \subset \mathfrak{p}$.*

Proof. We can assume that $\mathfrak{a} = (x)$ is principal. The set of ideals $\mathfrak{b}$ of A containing x satisfies the descending chain condition. Indeed, if $\mathfrak{b}_1 \supset \mathfrak{b}_2 \supset \dots \supset (x)$ then the chain $\mathfrak{b}_1^{-1} \subset \mathfrak{b}_2^{-1} \subset \dots \subset x^{-1}A$ is eventually stationary, since A is Noetherian. This implies the same for the original chain.

Suppose therefore that there are infinitely many distinct primes $\mathfrak{p}_1, \mathfrak{p}_2, \dots$ containing x . The sequence $\mathfrak{p}_1, \mathfrak{p}_1 \cap \mathfrak{p}_2, \dots$ is eventually stationary; suppose that it becomes stationary at step k . Then we have

$$\mathfrak{p}_{k+1} \supset \mathfrak{p}_1 \cap \dots \cap \mathfrak{p}_{k+1} = \mathfrak{p}_1 \cap \dots \cap \mathfrak{p}_k \supset \mathfrak{p}_1 \dots \mathfrak{p}_k.$$

We can find for each $i = 1, \dots, k$ an element $x_i \in \mathfrak{p}_i - \mathfrak{p}_{k+1}$. Then the element $x_1 \dots x_k \in \mathfrak{p}_1 \dots \mathfrak{p}_k$ must lie in $\mathfrak{p}_{k+1}$. Since the ideal $\mathfrak{p}_{k+1}$ is prime, this forces some x_i to lie in $\mathfrak{p}_{k+1}$, a contradiction. $\square$

If A is a Dedekind domain, then for each non-zero prime $\mathfrak{p} \subset A$, we have the valuation $v_{\mathfrak{p}} : K^{\times} \rightarrow \mathbb{Z}$ which is associated to the DVR $A_{\mathfrak{p}}$. If $\mathfrak{a} \subset K$ is a non-zero fractional ideal, then we have $\mathfrak{a}_{\mathfrak{p}} = aA_{\mathfrak{p}}$ for some $a \in K^{\times}$, and we define $v_{\mathfrak{p}}(\mathfrak{a}) = v_{\mathfrak{p}}(a)$. This defines a surjective homomorphism $v_{\mathfrak{p}} : \text{Div } A \rightarrow \mathbb{Z}$ (as $v_{\mathfrak{p}}(\mathfrak{p}) = 1$).

Proposition 1.16. *Let A be a Dedekind domain, and let $\mathfrak{a} \subset K$ be a non-zero fractional ideal.*

1. *We have $v_{\mathfrak{p}}(\mathfrak{a}) = 0$ for all but finitely many $\mathfrak{p}$.*
2. *The map $\oplus_{\mathfrak{p}} v_{\mathfrak{p}} : \text{Div } A \rightarrow \oplus_{\mathfrak{p}} \mathbb{Z}$ is an isomorphism.*
3. *We have $\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{v_{\mathfrak{p}}(\mathfrak{a})}$, where the product has only finitely many terms.*

Proof. We recall that if $\mathfrak{a}, \mathfrak{b} \subset K$ are fractional ideals, then $\mathfrak{a} \subset \mathfrak{b}$ if and only if $\mathfrak{a}_{\mathfrak{p}} \subset \mathfrak{b}_{\mathfrak{p}}$ for all $\mathfrak{p}$. If $\mathfrak{p} \neq \mathfrak{q}$ are non-zero prime ideals of A , then $\mathfrak{p}_{\mathfrak{q}} = A_{\mathfrak{q}}$. It follows that if $\mathfrak{a} \subset A$ is a non-zero ideal, then $\mathfrak{a} \subset \mathfrak{p}$ if and only if $\mathfrak{a}_{\mathfrak{p}} \subset \mathfrak{p}_{\mathfrak{p}}$, if and only if $v_{\mathfrak{p}}(\mathfrak{a}) > 0$. By the previous lemma, this can be true for only finitely many primes $\mathfrak{p}$. If $\mathfrak{a} \subset K$ is a non-zero fractional ideal, then we can find $x \in A - \{0\}$ such that $x\mathfrak{a} \subset A$ is an ideal. Then $v_{\mathfrak{p}}(x) = 0$ for all but finitely many $\mathfrak{p}$, and $v_{\mathfrak{p}}(x\mathfrak{a}) = 0$ for all but finitely many $\mathfrak{p}$. Since $v_{\mathfrak{p}}(x\mathfrak{a}) = v_{\mathfrak{p}}(x) + v_{\mathfrak{p}}(\mathfrak{a})$ for all non-zero prime ideals of A , this shows that $v_{\mathfrak{p}}(\mathfrak{a}) = 0$ for all but finitely many $\mathfrak{p}$.

The two fractional ideals $\mathfrak{a}$ and $\prod_{\mathfrak{p}} \mathfrak{p}^{v_{\mathfrak{p}}(\mathfrak{a})}$ have the same localizations, so are the same. $\square$

Corollary 1.17. *Let A be a Dedekind domain. Then every non-zero ideal $\mathfrak{a} \subset A$ admits a unique expression $\mathfrak{a} = \prod_{i=1}^n \mathfrak{p}_i^{a_i}$, where the $\mathfrak{p}_i$ are pairwise distinct prime ideals of A . This expression is uniquely determined up to re-ordering of terms.*

2 Complete discrete valuation rings

Definition 2.1. *Suppose given for each $i = 1, 2, \dots$ a group A_i and a homomorphism $f_i : A_{i+1} \rightarrow A_i$. The inverse limit is by definition*

$$\varprojlim_i A_i = \{(a_i) \in \prod_{i=1}^{\infty} A_i \mid \forall i \geq 1, f_i(a_{i+1}) = a_i\}.$$

It is a group (in fact a subgroup of $\prod_{i=1}^{\infty} A_i$). If the A_i are all abelian groups (resp. rings) and the f_i are homomorphisms of abelian groups (resp. rings) then it is naturally an abelian group (resp. ring).

Definition 2.2. *Let A be a discrete valuation ring. We say that A is complete if the natural map $A \rightarrow \varprojlim_i A/\mathfrak{m}_A^i$ is an isomorphism.*

We note that the kernel of the map $A \rightarrow \varprojlim_i A/\mathfrak{m}_A^i$ is $\cap_i \mathfrak{m}_A^i = 0$ (e.g. by Nakayama's lemma). The terminology is justified by the following lemma.

Lemma 2.3. *Let A be a discrete valuation ring with fraction field K and valuation $v : K^\times \rightarrow \mathbb{Z}$. Then the following are equivalent:*

1. A is complete.
2. A is complete as a metric space with respect to the metric $d(x, y)$ defined by $d(x, y) = 2^{-v(x-y)}$ if $x \neq y$ and $d(x, y) = 0$ otherwise.
3. K is a complete metric space with respect to the metric $d(x, y)$ given by the same formula.

Proof. Let $\pi \in A$ be a uniformizer. Suppose first that A is complete, and let $(x_i)_{i=0}^\infty$ be a Cauchy sequence in A . This is equivalent to the following condition: for all $M > 0$, there exists $N(M) > 0$ such that $x_i \equiv x_j \pmod{(\pi^M)}$ for all $i, j \geq N * M$. In particular, we can define an element $(a_i)_{i \geq 1} \in \varprojlim_i A/(\pi^i)$ by the formula $a_i = x_{N(i)}$. Since A is complete, there exists a unique $a \in A$ such that $a \equiv a_i \pmod{(\pi^i)}$ for each $i \geq 1$. Consequently, $a \equiv x_N \pmod{(\pi^i)}$ for all $N \geq N(i)$, hence $d(a, x_N) \leq 2^{-i}$ for all $N \geq N(i)$. It follows that $x_i \rightarrow a$ as $i \rightarrow \infty$.

Now suppose that A is complete as a metric space. We will show that A is a complete DVR. Let $(a_i)_{i \geq 1} \in \varprojlim_i A/(\pi^i)$, and choose elements $x_i \in A$ such that $a_i = x_i \pmod{(\pi^i)}$.

Then for any $j \geq i$, we have $x_j \equiv x_i \pmod{(\pi^i)}$ (by definition of the inverse limit), hence $d(x_j, x_i) \leq 2^{-i}$. Thus $(x_i)_{i \geq 1}$ is a Cauchy sequence, which admits a limit $a \in A$. We claim that the map $A \rightarrow \varprojlim_i A/(\pi^i)$ sends a to $(a_i)_{i \geq 1}$. It is enough to show that $a \equiv x_i \pmod{(\pi^i)}$

for each $i \geq 1$. Since $x_j \rightarrow a$, there exists $N(i)$ such that for each $j \geq N(i)$, $x_j \equiv a \pmod{(\pi^i)}$. Choosing $j \geq \max(N(i), i)$, we have $x_i \equiv x_j \equiv a \pmod{(\pi^i)}$, the first congruence by definition of the inverse limit, the second by definition of $N(i)$. This shows that A is complete.

Now suppose that K is a complete metric space. A is a closed subspace, since it is given by the formula $A = \{x \in K \mid d(0, x) \leq 1\}$. Therefore A is a complete metric space.

Now suppose that A is complete as a metric space. Note that for any $i \in \mathbb{Z}$ and $x, y \in K$, we have $d(\pi^i x, \pi^i y) = 2^{-i} d(x, y)$. If $(x_i)_{i \geq 1}$ is a Cauchy sequence in K , then there exists N such that for all $i, j \leq N$, $d(x_i, x_j) \leq 1$, hence $x_i - x_j \in A$. Choosing M such that $\pi^M x_i \in A$ for all $i = 1, \dots, N$, we find that $\pi^M x_i \in A$ for all $i \geq 1$. Therefore $(\pi^M x_i)_{i \geq 1}$ is a Cauchy sequence in A , so tends to a limit $\pi^M x$ in A . Then $x_i \rightarrow \pi^{-M} x$, and so K is complete. $\square$

When we speak on the topology of a complete DVR (or its field of fractions), we always mean the topology induced by this metric.

Proposition 2.4. *Let A be a DVR with uniformizer π , and define $\hat{A} = \varprojlim_i A/\mathfrak{m}_A^i$. Then;*

1. $\hat{A}$ is a complete DVR with uniformizer π .
2. For each $i \geq 0$, the natural map $A/\mathfrak{m}_A^i \rightarrow \hat{A}/\mathfrak{m}_{\hat{A}}^i$ is an isomorphism.
3. Let $X \subset A$ be a set of representatives for the residue field k_A containing 0. Then every element $x \in \hat{A}$ admits a unique π -adic expansion (a convergent infinite sum)

$$x = \sum_{i=0}^{\infty} a_i \pi^i$$

with $a_i \in X$ for each $i \geq 0$, and conversely every such expansion defines an element of $\hat{A}$.

Proof. We observe that the map $A/\pi \rightarrow \pi^i A/\pi^{i+1}A$ given by multiplication by π^i is an isomorphism. It follows that each element of the ring A/π^{i+1} admits a unique expression of the form $\sum_{j=0}^i a_j \pi^j$ with $a_j \in X$. The map $A/\pi^{i+1} \rightarrow A/\pi^i$ corresponds to throwing away the term $a_i \pi^i$. Since an element of $\hat{A}$ is a compatible system of elements of the quotients A/π^{i+1} , we see that the elements of $\hat{A}$ are in bijection with the expressions $\sum_{j=0}^{\infty} a_j \pi^j$ with $a_j \in X$. Since we have not yet proved that $\hat{A}$ is a DVR (and in particular, have not yet endowed $\hat{A}$ with a metric), this expression must be interpreted as follows: it corresponds to the tuple $(\sum_{j=0}^{i-1} a_j \pi^j)_{i \geq 1} \in \varprojlim_i A/(\pi^i)$.

By definition, we have for each $i \geq 0$ a surjection $\hat{A} \rightarrow A/\pi^i A$. An element $x = \sum_j a_j \pi^j$ lies in the kernel if and only if all of its digits $a_0, \dots, a_{i-1}$ are 0. In this case, we have $x = \pi^i \sum_{j=0}^{\infty} a_{j+i} \pi^j$, showing that the kernel of this map is $\pi^i \hat{A}$, and hence we have an isomorphism $\hat{A}/\pi^i \hat{A} \cong A/\pi^i A$.

The natural map $A \rightarrow \hat{A}$ is injective. Every element $x = \sum_{j=0}^{\infty} a_j \pi^j$ with $a_0 \neq 0$ is a unit. Indeed, we write $x = a_0(1 - \pi y)$ where $y = -\sum_{j=0}^{\infty} a_0^{-1} a_{j+1} \pi^j \in \hat{A}$. The element a_0 is a unit in A , hence in $\hat{A}$, and $(1 - \pi y)$ has inverse given by

$$1 + \pi y + (\pi y)^2 + (\pi y)^3 + \dots$$

This shows that every non-zero element of $\hat{A}$ has a unique expression as $u\pi^n$ with $u \in \hat{A}^\times$ and $n \geq 0$, and hence that $\hat{A}$ is a DVR. We have

$$\hat{A} = \varprojlim_i A/\pi^i A \cong \varprojlim_i \hat{A}/\pi^i \hat{A}$$

by the second part of the proposition, showing that $\hat{A}$ is complete. Finally, we observe that if $x \in \hat{A}$ corresponds to the expression $\sum_{j=0}^{\infty} a_j \pi^j$, then in fact this infinite sum converges in $\hat{A}$ and its limit equals x . $\square$

We refer to the ring $\hat{A}$ as the completion of A ; the proof of the proposition shows that if A is complete, then $A \cong \hat{A}$.

Definition 2.5. Let p be a prime. The ring of p -adic integers $\mathbb{Z}_p$ is by definition the completion of $\mathbb{Z}_{(p)}$. The field of p -adic numbers $\mathbb{Q}_p$ is its fraction field.

Thus $\mathbb{Z}_p$ is a complete DVR with residue field $\mathbb{Z}_p/(p) \cong \mathbb{Z}/p\mathbb{Z} = \mathbb{F}_p$. Every element of $\mathbb{Z}_p$ admits a unique expression

$$a_0 + a_1p + a_2p^2 + \dots$$

with $a_i \in \{0, \dots, p-1\}$. Multiplication and addition is done in the same way as for formal power series, except we now need to ‘carry’ digits.

The following lemma could be said to play the role of the intermediate value theorem in the p -adic setting. It is one example of a family of results which are referred to by the name ‘Hensel’s lemma’.

Lemma 2.6. Let A be a complete DVR with valuation v , and let $f(X) \in A[X]$ be a monic polynomial. Suppose there exists $x \in A$ such that $v(f(x)) > 2v(f'(x))$. Then there exists a unique $\alpha \in A$ such that $f(\alpha) = 0$ and $v(\alpha - x) > v(f'(x))$.

Proof. We use Newton’s approximation. Let $a_1 = x$, and define a sequence inductively by the formula

$$a_{n+1} = a_n - \frac{f(a_n)}{f'(a_n)}.$$

Let $t = v(f(a_1)/f'(a_1)^2) > 0$. We will show by induction on $n \geq 1$ that the following conditions are satisfied:

1. $v(a_n) \geq 0$.
2. $v(f'(a_n)) = v(f'(a_1))$.
3. $v(f(a_n)) \geq 2v(f'(a_1)) + 2^{n-1}t$.

These conditions hold for $n = 1$, by hypothesis. We now treat the induction step. Suppose the given conditions hold for $a_1, \dots, a_n$. We will show they hold for a_{n+1} . We have $v(a_{n+1}) \geq 0$ if and only if $f(a_n)/f'(a_n) \in A$. We have

$$v(f(a_n)) - v(f'(a_n)) \geq v(f'(a_n)) + 2^{n-1}t = v(f'(a_1)) + 2^{n-1}t > 0,$$

so the first condition holds. For the second, it is enough (by induction) to show that $v(f'(a_{n+1})) = v(f'(a_n))$. The element $f'(a_{n+1}) - f'(a_n)$ is divisible by $a_{n+1} - a_n$, so it is enough to show that $v(a_{n+1} - a_n) > v(f'(a_n))$, or equivalently that

$$v(f(a_n)/f'(a_n)) > v(f'(a_n)),$$

or even that

$$v(f(a_n)) > 2v(f'(a_1)).$$

Since $t > 0$ this is implied by the third condition (at step n).

Finally we show the third condition. Writing $f(X) = \sum_i a_i X^i$, we note that we have

$$f(X + Y) = \sum_i a_i (X + Y)^i = \sum_i a_i (X^i + iX^{i-1}Y + g_i(X, Y))$$

for some polynomial $g_i(X, Y) \in \mathbb{Z}[X, Y]$ depending only on i ; this is a consequence of the binomial theorem. Therefore we can write

$$f(X + Y) = f(X) + f'(X)Y + g(X, Y)Y^2$$

for some polynomial $g(X, Y) \in A[X, Y]$ (which now depends on $f(X)$). Setting $X = a_n$, $Y = -f(a_n)/f'(a_n)$, we obtain an expression

$$f(a_{n+1}) = g(a_n, -f(a_n)/f'(a_n))(f(a_n)/f'(a_n))^2,$$

hence

$$v(f(a_{n+1})) \geq 2v(f(a_n)) - 2v(f'(a_n)).$$

By induction, this gives

$$v(f(a_{n+1})) \geq 2(v(f'(a_n))) + 2^n t,$$

as desired. In particular, $f(a_n) \rightarrow 0$ as $n \rightarrow \infty$. Using the formula for a_{n+1} , we see that this implies that $(a_n)_{n \geq 1}$ is a Cauchy sequence, so there exists a limit $a \in A$. We must have $f(a) = 0$.

This shows the existence of a root. We must show that $v(a - x) > v(f'(x))$. Using the definition of a_n , we see that it is enough to show that $v(f(a_n)/f'(a_n)) > v(f'(a_n))$ for all $n \geq 1$. We have already seen this above.

We now establish uniqueness. If $b \in A$ is another root of f and $v(b - x) > v(f'(x))$, then we can write $b = a + h$ with $v(h) > v(f'(x))$. Using again the partial Taylor expansion for f , we find

$$0 = f(b) = f(a + h) = f(a) + hf'(a) + h^2g(a, h) = hf'(a) + h^2g(a, h).$$

If $h \neq 0$, we can cancel to obtain $f'(a) = -hg(a, h)$, hence $v(f'(a)) \geq v(h) > v(f'(x))$. This is a contradiction, showing that $h = 0$ and $a = b$, as desired. $\square$

Corollary 2.7. *Let A be a complete DVR with valuation v , and let $f(X) \in A[X]$ be a monic polynomial. Let $\bar{f}(X) \in k_A[X]$ be the reduction of $f(X)$ modulo $\mathfrak{m}_A$. Suppose there exists $x \in k_A$ such that $\bar{f}(x) = 0$ and $\bar{f}'(x) \neq 0$. Then there exists a unique $y \in A$ such that $f(y) = 0$ and $y \equiv x \pmod{\mathfrak{m}_A}$.*

Example 2.8. We can use this to understand which elements of $\mathbb{Q}_p$ are squares. Any non-zero element has a unique expression $p^n u$ with $u \in \mathbb{Z}_p^\times$, so it is equivalent to understand which elements of $\mathbb{Z}_p^\times$ are squares. A necessary condition is that $u \pmod{p} \in \mathbb{F}_p^\times$ is a square, so let us suppose this condition holds.

We apply Hensel's lemma to the polynomial $f(X) = X^2 - u$. Let v be an element of $\mathbb{Z}_p^\times$ such that $v^2 \equiv u \pmod{p}$. If p is odd, then $f'(v) = 2v$ is a p -adic unit, so Hensel's lemma shows that there is a unique $w \in \mathbb{Z}_p^\times$ such that $w^2 = u$.

If p is even, then a necessary condition is that $u \bmod 8 \in (\mathbb{Z}/8\mathbb{Z})^\times$ is a square, i.e. that $u \equiv 1 \bmod 8$. This being the case, we have $f(1) \equiv 0 \bmod 8$ and $f'(1) = 2$, hence $v(f(1)) \geq 3$ and $v(f'(1)) = 1$. Hensel's lemma shows in this case that there is a unique $w \in \mathbb{Z}_p^\times$ such that $w^2 = u$.

Example 2.9. The group homomorphism $\mathbb{Z}_p^\times \rightarrow \mathbb{F}_p^\times$ is surjective. We can use Hensel's lemma to construct a section to this map, i.e. a homomorphism $\tau : \mathbb{F}_p^\times \rightarrow \mathbb{Z}_p^\times$ such that $\tau(a) \bmod p = a$. Indeed, let $f(X) = X^p - X$. Then $f'(X) \equiv -1 \bmod p$, so the simple version of Hensel's lemma shows that for each $a \in \mathbb{F}_p$, there is a unique $b = \tau(a) \in \mathbb{Z}_p$ such that $b^p = b$ and $b \equiv a \bmod p$. The uniqueness implies that $\tau(aa') = \tau(a)\tau(a')$, hence τ is a group homomorphism. The map $a \mapsto \tau(a)$ is called the Teichmüller lift, and exists for any complete DVR with finite residue field. (If the residue field has q elements, use the polynomial $X^q - X$.)

We observe that the existence of τ implies that $\mathbb{Q}_p^\times$ contains all of the $(p-1)^{\text{th}}$ roots of unity. Here is a particular consequence.

Lemma 2.10. *Let q be a prime dividing $(p-1)$. Then $\mathbb{Q}_p$ has exactly $q+1$ isomorphism classes of Galois extensions of degree q .*

Proof. By Kummer theory, such extensions correspond to subgroups of $\mathbb{Q}_p^\times / (\mathbb{Q}_p^\times)^q$ of order q . We have an isomorphism

$$\begin{aligned} \mathbb{Z} \times (1 + p\mathbb{Z}_p) \times \mathbb{F}_p^\times &\cong \mathbb{Q}_p^\times \\ (n, u, a) &\mapsto p^n u \tau(a), \end{aligned}$$

hence an isomorphism

$$\mathbb{Q}_p^\times / (\mathbb{Q}_p^\times)^q \cong \mathbb{Z}/q\mathbb{Z} \times \mathbb{F}_p^\times / (\mathbb{F}_p^\times)^q.$$

Indeed, we have

$$1 + p\mathbb{Z}_p \cong \varprojlim_i \ker((\mathbb{Z}/p^i\mathbb{Z})^\times \rightarrow (\mathbb{Z}/p\mathbb{Z})^\times),$$

showing that if $n \in \mathbb{N}$ and $(n, p) = 1$ then the map $x \mapsto x^n$ is bijective on $1 + p\mathbb{Z}_p$ (it is an inverse limit of abelian groups of p -power order).

The group $\mathbb{F}_p^\times / (\mathbb{F}_p^\times)^q$ is cyclic of order q , so the lemma follows on noting that $(\mathbb{Z}/q\mathbb{Z})^2$ has exactly $q+1$ subgroups of order q . $\square$

If p is odd then $p-1$ is even, so we have:

Corollary 2.11. *If p is an odd prime, then $\mathbb{Q}_p$ has exactly 3 isomorphism classes of quadratic extensions.*

3 Extensions of Dedekind domains

Let A be a Dedekind domain with field of fractions K , and let E/K be a finite separable extension. An element $\gamma \in E$ is said to be integral over A if it satisfies an equation of the form

$$\gamma^n + a_1\gamma^{n-1} + \cdots + a_n = 0$$

with $a_1, \dots, a_n \in A$.

Lemma 3.1. *The following are equivalent:*

1. γ is integral over A .
2. $A[\gamma]$ is a finitely generated A -submodule of E .
3. There exists a non-zero $A[\gamma]$ -submodule $M \subset E$ which is finitely generated as an A -module.

Proof. 1 implies 2 because $1, \gamma, \dots, \gamma^{n-1}$ generate $A[\gamma]$. 2 trivially implies 3. To show 3 implies 1, let M be an $A[\gamma]$ -submodule of E which is finitely generated as an A -module. Then $\text{End}_A(M)$ is a finitely generated A -module which contains the elements $1, \gamma, \gamma^2, \dots$. Since A is Noetherian γ must satisfy a monic relation with coefficients in A . $\square$

Let B be the integral closure of A in E , i.e. the set of elements $b \in E$ satisfying an equation of the form

$$b^n + a_1 b^{n-1} + \dots + a_n = 0$$

with $a_1, \dots, a_n \in A$.

Lemma 3.2. *B is a ring, which is integrally closed in E .*

Proof. We first show that B is a ring. This means showing that if $b, b' \in B$, then $b + b'$ and bb' are integral over A . However, $A[b, b']$ is a finitely generated A -module which is stable under multiplication by $b + b'$ and bb' .

We next show B is integrally closed in E . Let $\gamma \in E$ be integral over B . Thus γ satisfies a relation $\gamma^n + b_1 \gamma^{n-1} + b_n = 0$ with $b_1, \dots, b_n \in B$. It follows that $A[b_1, b_2, \dots, b_n, \gamma]$ is a finitely generated A -module, hence that γ is integral over A , hence $\gamma \in B$. $\square$

A useful observation is that the formation of B commutes with localization. More precisely, let $S \subset A - \{0\}$ be a multiplicative subset. Then $S^{-1}A$ is a Dedekind domain with field of fractions K , and the integral closure of $S^{-1}A$ equals $S^{-1}B$. In particular, $(A - \{0\})^{-1}B$ equals the integral closure of $(A - \{0\})^{-1}A = K$, namely E . This shows that B always spans E as a K -vector space.

Lemma 3.3. *Let E/K be a finite separable extension of fields. Then the symmetric K -bilinear form $T : E \times E \rightarrow K$ given by the formula $T(x, y) = \text{tr}_{E/K}(xy)$ is non-degenerate.*

Proof. We must show that for any non-zero $x \in E$, there exists $y \in E$ such that $\text{tr}_{E/K}(xy) \neq 0$. It is enough to show that $\text{tr}_{E/K} \neq 0$ as a homomorphism $E \rightarrow K$. If L/K denotes the Galois closure of E , then there are exactly n distinct K -embeddings $\sigma_1, \dots, \sigma_n : E \hookrightarrow L$; and we have $\text{tr}_{E/K}(x) = \sigma_1(x) + \dots + \sigma_n(x)$. The lemma therefore follows from the fact that the embeddings $\sigma_1, \dots, \sigma_n$ are linearly independent over L (even as homomorphisms $E^\times \rightarrow L^\times$). $\square$

Proposition 3.4. *B is a finitely generated A -module. It is a Dedekind domain.*

Proof. Let $e_1, \dots, e_n$ be a K -basis for E . After multiplying through by elements of A , we can suppose that each $e_i \in B$. Let us note that if $z \in B$, then $\text{tr}_{E/K} z$ is integral over A and lies in K , hence lies in A . Indeed, let L/K denote the Galois closure of E/K , and let $\sigma_1, \dots, \sigma_n : E \rightarrow L$ be the distinct K -embeddings. Then $\text{tr}_{E/K}(z) = \sigma_1(z) + \dots + \sigma_n(z)$ lies in K (it is a trace) and is integral over A (it is a sum of elements of the integral closure of A in L). Since A is integrally closed in K , we find $\text{tr}_{E/K}(z) \in A$.

Let $f_1, \dots, f_n \in E$ be the dual K -basis with respect to the non-degenerate symmetric bilinear form T , so that $T(e_i, f_j) = \delta_{ij}$. We claim that $B \subset \sum_{i=1}^n Af_i$. Since A is Noetherian, this will show that B satisfies the ascending chain condition for A -submodules, hence for B -submodules, hence that B is a Noetherian ring. To see this, let $b \in B$ be a general element, and write $b = \sum_{i=1}^n a_i f_i$ with $a_i \in K$. Then for any $j = 1, \dots, n$ we have $T(b, e_j) = \text{tr}_{E/K}(be_j) = a_j \in A$.

Let $\mathfrak{q} \subset B$ be a non-zero prime ideal. To complete the proof, we must show that $\mathfrak{q}$ is a maximal ideal; equivalently, that $B/\mathfrak{q}$ is a field. Let $\mathfrak{p} = \mathfrak{q} \cap A$. Then $\mathfrak{p} \subset A$ is prime. It is also non-zero: if $b \in \mathfrak{q}$ is a non-zero element, then we can find an equation of minimal degree with $a_i \in A$:

$$b^n + a_1 b^{n-1} + \dots + a_n = 0.$$

Then $a_n \in \mathfrak{q} \cap A = \mathfrak{p}$ and $a_n \neq 0$, by minimality.

There is an injective homomorphism $A/\mathfrak{p} \hookrightarrow B/\mathfrak{q}$, $A/\mathfrak{p}$ is a field, $B/\mathfrak{q}$ is a domain, and $B/\mathfrak{q}$ is finite-dimensional as $A/\mathfrak{p}$ -vector space. This implies that $B/\mathfrak{q}$ is in fact a field. Indeed, let $\alpha \in B/\mathfrak{q}$ be non-zero, and consider the map $m_\alpha : B/\mathfrak{q} \rightarrow B/\mathfrak{q}$ given by the formula $m_\alpha(\beta) = \alpha\beta$. This is an $A/\mathfrak{p}$ -linear map, and it is injective because $B/\mathfrak{q}$ is a domain. Since an injective endomorphism of a finite-dimensional vector space is surjective, we see that there exists $\beta \in B/\mathfrak{q}$ such that $\alpha\beta = 1$, hence α is a unit and $B/\mathfrak{q}$ is a field. $\square$

Part of the above proof motivates the following definition:

Definition 3.5. If $\mathfrak{q} \subset B$ is a non-zero prime ideal and $\mathfrak{p} = \mathfrak{q} \cap A$, then we say that $\mathfrak{q}$ lies above $\mathfrak{p}$.

Lemma 3.6. Let $\mathfrak{q} \subset B$, $\mathfrak{p} \subset A$ be non-zero prime ideals. Then the following are equivalent:

1. $\mathfrak{q}$ lies above $\mathfrak{p}$.
2. $\mathfrak{q}$ appears in the prime factorization of $\mathfrak{p}B \subset B$ (in other notation, $v_{\mathfrak{q}}(\mathfrak{p}B) > 0$).

Proof. If $\mathfrak{q}$ lies above $\mathfrak{p}$ then $\mathfrak{p}B \subset \mathfrak{q}$, hence $\mathfrak{q}B_{\mathfrak{q}} \subset \mathfrak{q}B_{\mathfrak{q}}$, hence $v_{\mathfrak{q}}(\mathfrak{p}B) > 0$. Conversely, if $v_{\mathfrak{q}}(\mathfrak{p}B) > 0$ then $\mathfrak{p}B_{\mathfrak{q}} \subset \mathfrak{q}B_{\mathfrak{q}}$, hence (using $\mathfrak{q}B_{\mathfrak{q}} \cap B = \mathfrak{q}$) $\mathfrak{p} \subset \mathfrak{q}$, hence $\mathfrak{q} \cap A = \mathfrak{p}$ and $\mathfrak{q}$ lies above $\mathfrak{p}$. $\square$

Henceforth, if $\mathfrak{q}$ is a non-zero prime ideal in a Dedekind domain, then we write $k_{\mathfrak{q}}$ for its residue field.

Definition 3.7. Suppose that $\mathfrak{q} \subset B$ is a non-zero prime ideal, and let $\mathfrak{p} = \mathfrak{q} \cap A$. Then there is a natural embedding of fields $k_{\mathfrak{p}} \rightarrow k_{\mathfrak{q}}$. We define the ramification index $e_{\mathfrak{q}/\mathfrak{p}} = v_{\mathfrak{q}}(\mathfrak{p}B)$ and the residue degree $f_{\mathfrak{q}/\mathfrak{p}} = [B/\mathfrak{q} : A/\mathfrak{p}]$.

Proposition 3.8. *Let $\mathfrak{p} \subset A$ be a non-zero prime ideal, and let $n = [E : K]$. Then we have $n = \sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}/\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}}$, the sum running over all primes of B lying over $\mathfrak{p}$.*

Proof. Let $S = (A - \mathfrak{p})$. After replacing A by $S^{-1}A$ and B by $S^{-1}B$, we can assume that A is a DVR. (Note that $S^{-1}B$ is the integral closure of $S^{-1}A$, and if $\mathfrak{q} \cap S = \emptyset$ then replacing $\mathfrak{q}$ by $S^{-1}\mathfrak{q}$ does not change e or f .) Then B is a torsion-free finitely generated A -module, so is free of some finite rank. We have $(A - \{0\})^{-1}B = E$, so this rank is $\dim_K E = n$.

To prove the proposition, we will calculate this rank in another way. Reducing modulo $\mathfrak{p}$, we also have $n = \dim_{A/\mathfrak{p}} B/\mathfrak{q}B$. By the Chinese remainder theorem, there is an isomorphism

$$B/\mathfrak{p}B \cong \prod_{\mathfrak{q}|\mathfrak{p}} B/\mathfrak{q}^{e_{\mathfrak{q}/\mathfrak{p}}},$$

so we get $n = \sum_{\mathfrak{q}|\mathfrak{p}} \dim_{A/\mathfrak{p}} B/\mathfrak{q}^{e_{\mathfrak{q}/\mathfrak{p}}}$. There is a filtration $B \supset \mathfrak{q} \supset \mathfrak{q}^2 \supset \cdots \supset \mathfrak{q}^{e_{\mathfrak{q}/\mathfrak{p}}}$, and each $\mathfrak{q}^i/\mathfrak{q}^{i+1}$ is a $B/\mathfrak{q}$ -vector space of dimension 1, hence an $A/\mathfrak{p}$ -vector space of dimension $f_{\mathfrak{q}/\mathfrak{p}}$. We obtain

$$\dim_{A/\mathfrak{p}} B/\mathfrak{q}^{e_{\mathfrak{q}/\mathfrak{p}}} = e_{\mathfrak{q}/\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}},$$

which completes the proof. $\square$

It is useful to introduce the following language.

Definition 3.9. *Let $\mathfrak{p}$ be a non-zero prime ideal of A . We say that $\mathfrak{p}$ is unramified in B (or E) if for each prime ideal $\mathfrak{q}$ of B lying above $\mathfrak{p}$, the extension $k_{\mathfrak{q}}/k_{\mathfrak{p}}$ is separable and $f_{\mathfrak{q}/\mathfrak{p}} = 1$.*

We say that $\mathfrak{p}$ splits completely in B (or E) if for each prime ideal $\mathfrak{q}$ of B lying above $\mathfrak{p}$, we have $e_{\mathfrak{q}/\mathfrak{p}} = f_{\mathfrak{q}/\mathfrak{p}} = 1$. (Thus the number of such primes $\mathfrak{q}$ is $[E : K]$, by the proposition).

We now compute an example. If E is a number field, then we will write $\mathcal{O}_E$ for the integral closure of $\mathbb{Z}$ in E ; this is traditionally called the ring of integers of E . The most elementary family of number fields is given by the quadratic extensions $E/\mathbb{Q}$; each of these may be represented uniquely in the form $E = \mathbb{Q}(\sqrt{d})$, where $d \in \mathbb{Z}$ is a squarefree integer such that $d \not\equiv 0, 1$.

Let us first compute $\mathcal{O}_E$. We claim that $\mathcal{O}_E = \mathbb{Z}[\sqrt{d}]$ if $d \equiv 2, 3 \pmod{4}$, and $\mathcal{O}_E = \mathbb{Z}[(1 + \sqrt{d})/2]$ if $d \equiv 1 \pmod{4}$. The elements $\sqrt{d}$ and $(1 + \sqrt{d})/2$ satisfy the equations $X^2 - d = 0$ and $X^2 - X + (1 - d)/4 = 0$, respectively, showing that they are integral over $\mathbb{Z}$. Therefore what we need to show is that $\mathcal{O}_E$ can not be any larger than the claimed rings.

Let $\alpha \in \mathcal{O}_E$: then we can write $\alpha = a + b\sqrt{d}$ for some $a, b \in \mathbb{Q}$. We observe that $\text{Gal}(E/\mathbb{Q})$ acts on $\mathcal{O}_E$. Indeed, if $\gamma \in E$ satisfies an equation $\gamma + a_1\gamma^{n-1} + \cdots + a_n$ with $a_i \in \mathbb{Z}$, and $\sigma \in \text{Gal}(E/\mathbb{Q})$, then $\sigma(\gamma)$ satisfies the same equation. In particular, $\alpha' = a - b\sqrt{d}$ also lies in $\mathcal{O}_E$, hence $\alpha + \alpha' = 2a$ and $\alpha\alpha' = a^2 - db^2$ lie in $\mathcal{O}_E \cap \mathbb{Q} = \mathbb{Z}$.

We can therefore write $a = u/2$ with $u \in \mathbb{Z}$. Then $u^2 - 4db^2$ is an integer. Considering p -adic valuations for every prime p shows that $2b$ must also be an integer, so we can write $b = v/2$ with $v \in \mathbb{Z}$. Then $u^2 - dv^2$ must be in $4\mathbb{Z}$, hence $u^2 \equiv dv^2 \pmod{4}$.

The only squares modulo 4 are 0, 1. If $d \equiv 2, 3 \pmod{4}$ then we see that u, v must both be even, hence a, b are both integers and α lies in $\mathbb{Z}[\sqrt{d}]$. If $d \equiv 1 \pmod{4}$ then we see that u, v must have the same parity, and therefore that α lies in $\mathbb{Z}[(1 + \sqrt{d})/2]$.

This completes the computation of $\mathcal{O}_E$. We now show how to factorize $p\mathcal{O}_E$ when p is a prime number. We first consider the case where p is odd. In this case it is enough to factorize $p\mathcal{O}_E[1/2]$ (localization at the multiplicative subset $\{1, 2, 2^2, \dots\}$). This ring equals $\mathbb{Z}[1/2, \sqrt{d}]$, independent of the value of d modulo 4.

We compute $\mathbb{Z}[1/2, \sqrt{d}]/(p) = \mathbb{Z}[1/2, X]/(p, X^2 - d) = \mathbb{F}_p[X]/(X^2 - d)$. There are 3 possibilities: if d is a quadratic residue modulo p then $X^2 - d$ splits into linear factors, showing that $\mathcal{O}_E/(p)$ is a product of fields and p splits completely in E . If d is a quadratic non-residue modulo p then $X^2 - d$ is irreducible, showing that $\mathcal{O}_E/(p)$ is a quadratic extension of $\mathbb{F}_p$, and hence that p is unramified in $\mathcal{O}_E$ and does not split. Finally if p divides E then $\mathcal{O}_E/(p) = \mathbb{F}_p[X]/(X^2)$ has non-trivial nilpotent elements, which implies that p must ramify in $\mathcal{O}_E$.

We now treat the case $p = 2$. If $d \equiv 2, 3 \pmod{4}$ then we find $\mathcal{O}_E/(2) = \mathbb{Z}[\sqrt{d}]/(2) = \mathbb{F}_2/(X^2 - d) = \mathbb{F}_2/((X - d)^2)$, which shows that 2 ramifies in E . If $d \equiv 1 \pmod{4}$ then we see that $\mathcal{O}_E/(2) = \mathbb{F}_2[X]/(X^2 - X + (1 - d)/4)$, showing that 2 splits if $(1 - d)/4$ is even and is unramified and does not split if $(1 - d)/4$ is odd.

We now return to our general discussion of the integral closure of A in an extension E/K , and suppose moreover that E/K is Galois. The same argument as in the quadratic case shows that $G = \text{Gal}(E/K)$ acts on B .

Proposition 3.10. *Suppose that the extension E/K is Galois, and let $\mathfrak{p} \subset A$ be a non-zero prime ideal. Then:*

1. *The group G acts transitively on the set of primes of B lying above $\mathfrak{p}$.*
2. *Let $\mathfrak{q}$ be a prime lying over $\mathfrak{p}$, and let $\sigma \in G$. Then $f_{\sigma(\mathfrak{q})/\mathfrak{p}} = f_{\mathfrak{q}/\mathfrak{p}}$ and $e_{\sigma(\mathfrak{q})/\mathfrak{p}} = e_{\mathfrak{q}/\mathfrak{p}}$.*
3. *For any prime $\mathfrak{q}$ of B lying over $\mathfrak{p}$, we have $[E : K] = e_{\mathfrak{q}/\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}} g_{\mathfrak{q}/\mathfrak{p}}$, where $g_{\mathfrak{q}/\mathfrak{p}}$ is the number of distinct primes of B lying over $\mathfrak{p}$.*

Proof. We first explain why G acts on the set of prime ideals of B lying above $\mathfrak{p}$. If $\sigma \in G$ then the restriction of σ to B is a ring automorphism of B that acts as the identity on A . Therefore if $\mathfrak{q} \subset B$ is a non-zero prime ideal, then $\sigma(\mathfrak{q}) \subset B$ is also a non-zero prime ideal, and we have $\sigma(\mathfrak{q}) \cap A = \mathfrak{q} \cap A$.

To show that the action is transitive, we can first localize to replace A by $A_{\mathfrak{p}}$ and assume that A is a DVR. Let $\mathfrak{q}, \mathfrak{q}'$ be distinct primes of B lying above $\mathfrak{p}$, and let $\pi \in B$ be a generator of $\mathfrak{q}$. Then $N_{E/K}(\pi) = \prod_{\sigma \in G} \sigma(\pi) \in \mathfrak{q} \cap A = \mathfrak{p} \subset \mathfrak{q}'$. Since $\mathfrak{q}'$ is prime there exists $\sigma \in G$ such that $\sigma(\pi) \in \mathfrak{q}'$, hence $\sigma(\mathfrak{q}) \subset \mathfrak{q}'$, hence $\sigma(\mathfrak{q}) = \mathfrak{q}'$, as desired.

This shows the first part of the proposition. The second follows from the definitions. The third follows from the previous proposition. $\square$

If the extension E/K is Galois and $\mathfrak{q}$ is a non-zero prime of B lying above the prime $\mathfrak{p}$ of A , then we define $D_{\mathfrak{q}/\mathfrak{p}} = \text{Stab}_G(\mathfrak{q})$, and call it the decomposition group at the prime $\mathfrak{q}$.

Proposition 3.11. *Suppose that the extension E/K is Galois, and let $\mathfrak{q}$ be a prime of B lying above the prime $\mathfrak{p}$ of A such that the corresponding extension $k_{\mathfrak{q}}/k_{\mathfrak{p}}$ of residue fields is separable. Then it is Galois, and the induced map $D_{\mathfrak{q}/\mathfrak{p}} \rightarrow \text{Gal}(k_{\mathfrak{q}}/k_{\mathfrak{p}})$ is surjective.*

Proof. Let $\bar{x} \in k_q$, and let $x \in B$ be any element with $\bar{x} = x \bmod \mathfrak{q}$. Let $f(X) = \prod_{\sigma \in G} (X - \sigma(x)) \in A[X]$. Let $\bar{f}(X) = f(X) \bmod \mathfrak{p} \in k_p[X]$. By construction, the polynomial $\bar{f}(X)$ has $\bar{x}$ as a root and splits into linear factors in $k_q[X]$. Since $\bar{x}$ was arbitrary, this shows that k_q is normal, hence Galois over k_p .

To show that the map $D_{q/p} \rightarrow \text{Gal}(k_q/k_p)$ is surjective, let $\bar{x} \in k_q$ be a primitive element; this exists since the extension k_q/k_p is separable. We can assume that $\bar{x}$ is not 0 (since otherwise $k_q = k_p$ and the proposition is trivial). By the Chinese remainder theorem, we can choose an element $x \in B$ lifting $\bar{x}$ such that $x \in \mathfrak{q}'$ if $\mathfrak{q}' \neq \mathfrak{q}$ is any other prime of B lying over the prime $\mathfrak{p}$. Form the polynomial $f(X)$ as before; we now have $\bar{f}(X) = X^d \prod_{\sigma \in D_{q/p}} (X - \sigma(\bar{x}))$ for some $d \geq 0$.

If $\tau \in \text{Gal}(k_q/k_p)$, then $\tau(\bar{x})$ is a non-zero root of $\bar{f}(X)$, hence there exists $\sigma \in D_{q/p}$ such that $\sigma(\bar{x}) = \tau(\bar{x})$. Since $\bar{x}$ is a primitive element, this shows that the image of σ in $\text{Gal}(k_q/k_p)$ equals τ , as desired. $\square$

Definition 3.12. In the situation of the proposition, we call $I_{q/p} = \ker(D_{q/p} \rightarrow \text{Gal}(k_q/k_p))$ the inertia group at the prime $\mathfrak{q}$.

Observe that we have $\#I_{q/p} = e_{q/p}$; in particular, the inertia group is trivial if and only if the prime $\mathfrak{q}$ is unramified over $\mathfrak{p}$.

The above formalism is particularly pleasant when E/K is an extension of number fields. Indeed, if $\mathfrak{p} \subset \mathcal{O}_K$ is a non-zero prime ideal, then $\mathcal{O}_K/\mathfrak{p}$ is a finite field. Every extension of finite fields is separable, so if E/K is Galois we always have a surjective homomorphism $D_{q/p} \rightarrow \text{Gal}(k_q/k_p)$. Moreover, $\text{Gal}(k_q/k_p)$ is cyclic, with a canonical generator given by the Frobenius automorphism $x \mapsto x^{|k_p|}$. When $\mathfrak{p}$ is unramified in E , we therefore have a well-defined element $F_{q/p} \in \text{Gal}(E/K)$, called the Frobenius automorphism at $\mathfrak{q}$: it is the unique element of $D_{q/p}$ which induces the Frobenius automorphism on the residue field extension k_q/k_p .

This allows us to prove an important computational result from the Part II Galois Theory course. Let $f(X) = X^n + a_1X^{n-1} + \dots + a_n \in \mathbb{Z}[X]$ be an irreducible polynomial, and suppose that $E/\mathbb{Q}$ is the splitting field of $f(X)$; let $\alpha_1, \dots, \alpha_n \in E$ be the roots of $f(X)$. Then we can identify $\text{Gal}(E/\mathbb{Q})$ with a subgroup of the symmetric group on n letters (via its action by permutations of $\alpha_1, \dots, \alpha_n$).

The result states that if p is a prime such that $\bar{f}(X) = f(X) \bmod p$ factors as $\bar{f}(X) = \prod_{i=1}^k \bar{f}_i(X)$, where $\bar{f}_1(X), \dots, \bar{f}_k(X)$ are distinct irreducible polynomials in $\mathbb{F}_p[X]$, then $\text{Gal}(E/\mathbb{Q})$ contains an element in S_n of cycle type $(\deg \bar{f}_1) \dots (\deg \bar{f}_k)$.

This is the Frobenius element $F_{q/p}$! Indeed, let $\mathfrak{q}$ be a prime ideal of $\mathcal{O}_E$ lying above p , and let $\bar{\alpha}_1, \dots, \bar{\alpha}_n \in k_q$ be the images of the roots of $f(X)$ under reduction modulo $\mathfrak{q}$. Since $\bar{f}(X)$ has no repeated factors, it has no repeated roots, and $\bar{\alpha}_1, \dots, \bar{\alpha}_n$ are the distinct roots of $\bar{f}(X)$ in k_q . In particular, the map $D_{q/p} \rightarrow \text{Gal}(k_q/k_p)$ is injective, so $\mathfrak{p}$ is unramified in E .

The irreducible factors of $\bar{f}(X)$ in $k_p[X]$ correspond to orbits for $\text{Gal}(k_q/k_p)$ on $\bar{\alpha}_1, \dots, \bar{\alpha}_n$. This says precisely that the cycle type of $F_{q/p}$, as a permutation of $\alpha_1, \dots, \alpha_n$, equals $(\deg \bar{f}_1) \dots (\deg \bar{f}_k)$.

The above discussion shows how one can understand the decomposition group in the unramified situation. In general, the right tool is passage to completion. This is explained in the following proposition.

Proposition 3.13. *Let A be a Dedekind domain with field of fractions K , let E/K be a finite separable extension, let B be the integral closure of A in E , and let $\mathfrak{q}$ be a non-zero prime ideal of B lying above the prime $\mathfrak{p}$ of A . Then:*

1. *There is a canonical embedding of DVRs $\widehat{A}_{\mathfrak{p}} \rightarrow \widehat{B}_{\mathfrak{q}}$ extending the embedding $A \rightarrow B$.*
2. *Let $E_{\mathfrak{q}}, K_{\mathfrak{p}}$ denote the fields of fractions of $\widehat{B}_{\mathfrak{q}}$ and $\widehat{A}_{\mathfrak{p}}$, respectively. Then $E_{\mathfrak{q}}/K_{\mathfrak{p}}$ is a finite separable extension, equal to the compositum $K_{\mathfrak{p}} \cdot E$, and $\widehat{B}_{\mathfrak{q}}$ is the integral closure of $K_{\mathfrak{p}}$ in $E_{\mathfrak{q}}$.*
3. *We have $e_{\widehat{B}_{\mathfrak{q}}/\widehat{A}_{\mathfrak{p}}} = e_{\mathfrak{q}/\mathfrak{p}}$, $f_{\widehat{B}_{\mathfrak{q}}/\widehat{A}_{\mathfrak{p}}} = f_{\mathfrak{q}/\mathfrak{p}}$, and $[E_{\mathfrak{q}} : K_{\mathfrak{p}}] = e_{\mathfrak{q}/\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}}$.*
4. *Suppose further that the extension E/K is Galois. Then $E_{\mathfrak{q}}/K_{\mathfrak{p}}$ is Galois and the natural map $D_{\mathfrak{q}/\mathfrak{p}} \rightarrow \text{Gal}(E_{\mathfrak{q}}/K_{\mathfrak{p}})$ given by passage to completion is an isomorphism.*

Proof. We first note that we have $\widehat{A}_{\mathfrak{p}} \cong \varprojlim_i A/\mathfrak{p}^i$, and similarly for $\widehat{B}_{\mathfrak{q}}$. Indeed, it suffices to note that the natural map $A/\mathfrak{p}^i \rightarrow A_{\mathfrak{p}}/\mathfrak{p}_{\mathfrak{p}}^i$ is an isomorphism for any $i \geq 1$. Also, we are free to replace A by its localization $A_{\mathfrak{p}}$. We can therefore assume that A is a DVR, and hence that B is a finite free A -module.

We have natural maps $A/\mathfrak{p}^i \rightarrow B/\mathfrak{q}^i$ for each $i \geq 1$, and passage to the inverse limit gives a map $\widehat{A}_{\mathfrak{p}} \rightarrow \widehat{B}_{\mathfrak{q}}$. It is injective because the map $A \rightarrow B$ is injective. If we write $\mathfrak{q} = \mathfrak{q}_1, \dots, \mathfrak{q}_r$ for the distinct primes of B above $\mathfrak{p}$, then there is an isomorphism (by the Chinese Remainder Theorem)

$$\varprojlim_i B/\mathfrak{p}^i B \cong \prod_{i=1}^r \widehat{B}_{\mathfrak{q}_i}.$$

Since B is a finite free A -module, $\varprojlim_i B/\mathfrak{p}^i B$ is a finite free $\widehat{A}_{\mathfrak{p}}$ -module. We conclude that the quotient $\widehat{B}_{\mathfrak{q}}$ is a finite free $\widehat{A}_{\mathfrak{p}}$ -module. In particular, $\widehat{B}_{\mathfrak{q}}$ is integral over $\widehat{A}_{\mathfrak{p}}$, hence is the integral closure of $\widehat{A}_{\mathfrak{p}}$ in $E_{\mathfrak{q}}$.

The map $B/\mathfrak{p}B \rightarrow \widehat{B}_{\mathfrak{q}}/\mathfrak{p}\widehat{B}_{\mathfrak{q}}$ is surjective (for example, by considering π -adic expansions), so Nakayama's lemma shows that B generates $\widehat{B}_{\mathfrak{q}}$ as an $\widehat{A}_{\mathfrak{p}}$ -module. Passing to fraction fields, we see that E spans $E_{\mathfrak{q}}$ as a $K_{\mathfrak{p}}$ -vector space. In particular, $E_{\mathfrak{q}}$ is equal to the compositum of its subfields E and $K_{\mathfrak{p}}$. This shows $E_{\mathfrak{q}}/K_{\mathfrak{p}}$ is separable (it is obtained by adjoining the roots of a separable polynomial) and Galois if E/K is.

So far we have proved the first two parts of the proposition. For the third, we observe that $\widehat{A}_{\mathfrak{p}}/\mathfrak{p} = A/\mathfrak{p}$, and similarly for $\widehat{B}_{\mathfrak{q}}$. In other words, completion does not change residue fields, so in particular the residue degree f is unchanged. On the other hand we have $\mathfrak{p}\widehat{B}_{\mathfrak{q}} = \prod_{i=1}^r \mathfrak{q}_i^{e_{\mathfrak{q}_i/\mathfrak{p}}} \widehat{B}_{\mathfrak{q}} = \mathfrak{q}^{e_{\mathfrak{q}/\mathfrak{p}}} \widehat{B}_{\mathfrak{q}}$, since $\mathfrak{q}_i \widehat{B}_{\mathfrak{q}}$ is the unit ideal if $i > 1$. This shows the third part.

Finally, we suppose that E/K is Galois and prove the fourth part. If $\sigma \in D_{\mathfrak{q}/\mathfrak{p}}$, then $\sigma(\mathfrak{q}) = \mathfrak{q}$, hence $\sigma(\mathfrak{q}^i) = \mathfrak{q}^i$ for any $i \geq 1$. It follows that σ extends, by passage to inverse limit, to an automorphism of $\widehat{B}_{\mathfrak{q}}$ which fixes $\widehat{A}_{\mathfrak{p}}$. We obtain a homomorphism $D_{\mathfrak{q}/\mathfrak{p}} \rightarrow \text{Gal}(E_{\mathfrak{q}}/K_{\mathfrak{p}})$. It is injective because $E \subset E_{\mathfrak{q}}$. It is surjective because the source and target have the same cardinality (namely $e_{\mathfrak{q}/\mathfrak{p}}f_{\mathfrak{q}/\mathfrak{p}}$). $\square$

Corollary 3.14. *Let A be a Dedekind domain with field of fractions K , let E/K be a finite separable extension, let B be the integral closure of A in E , and let $\mathfrak{p}$ be a non-zero prime ideal of A . Let $\alpha \in E$ be such that $E = K(\alpha)$, and let $f(X) \in K[X]$ denote the minimal polynomial of α . Then the following two sets are in canonical bijection:*

1. *The prime ideals of B lying above the prime ideal $\mathfrak{p}$*
2. *The irreducible factors of $f(X)$ in $K_{\mathfrak{p}}[X]$.*

The correspondence is given as follows: a prime ideal $\mathfrak{q} \subset B$ lying above $\mathfrak{p}$ corresponds to the unique irreducible factor $g(X)$ of $f(X)$ in $K_{\mathfrak{p}}[X]$ such that $g(\alpha) = 0$ in $E_{\mathfrak{q}}$.

Proof. Let L/K be the Galois closure of E/K , and let $G = \text{Gal}(L/K)$, $H = \text{Gal}(L/E)$. Let C denote the integral closure of A in L , and let $\mathfrak{r}$ be a fixed prime ideal of C above $\mathfrak{p}$. We first note that there is an isomorphism of G -sets (i.e. sets with left G -action) between G/H and the set of roots of $f(X)$ in L , hence an isomorphism of $D_{\mathfrak{r}/\mathfrak{p}} = \text{Gal}(L_{\mathfrak{r}}/K_{\mathfrak{p}})$ -sets between G/H and the set of roots of $f(X)$ in $L_{\mathfrak{r}}$. It follows that the irreducible factors of $f(X)$ in $K_{\mathfrak{p}}[X]$ are in bijection with the set of orbits of $D_{\mathfrak{r}/\mathfrak{p}}$ on G/H . This set of orbits is identified with the double quotient $D_{\mathfrak{r}/\mathfrak{p}} \backslash G/H$, and element $\sigma \in G$ corresponding to the unique irreducible factor $g(X)$ of $f(X)$ in $K_{\mathfrak{p}}[X]$ such that $g(\sigma(\alpha)) = 0$ in $L_{\mathfrak{r}}$.

On the other hand, there is an isomorphism of G -sets between $G/D_{\mathfrak{r}/\mathfrak{p}}$ and the set of prime ideals of C lying above $\mathfrak{p}$, because G acts transitively with stabilizer $D_{\mathfrak{r}/\mathfrak{p}}$. The set of prime ideals of B lying above $\mathfrak{p}$ is in bijection with the set of H -orbits, because H acts transitively on the set of prime ideals of C above a given prime ideal of B . We conclude that the set of prime ideals of B lying above $\mathfrak{p}$ is in bijection with the double quotient $H \backslash G/D_{\mathfrak{r}/\mathfrak{p}}$, an element $\sigma \in G$ determining the prime ideal $\sigma(\mathfrak{r}) \cap B$.

These two double quotients are in bijection via the map $D_{\mathfrak{r}/\mathfrak{p}}\sigma H \mapsto H\sigma^{-1}D_{\mathfrak{r}/\mathfrak{p}}$. This bijection can be written explicitly as follows: if $g(X)$ is an irreducible factor of $f(X)$ in $K_{\mathfrak{p}}[X]$, we choose $\sigma \in G$ such that $g(\sigma(\alpha)) = 0$ in $L_{\mathfrak{r}}$. Then we take the prime ideal $\sigma^{-1}(\mathfrak{r}) \cap B$ of B .

To complete the proof, we must show that this bijection coincides with the one given in the statement of the corollary. Equivalently, we must show that if $\sigma \in G$, then $g(\sigma(\alpha)) = 0$ in $L_{\mathfrak{r}}$ if and only if $g(\alpha) = 0$ in $E_{\sigma^{-1}(\mathfrak{r}) \cap B}$. To see this, we note that σ^{-1} determines an isomorphism $L_{\mathfrak{r}} \rightarrow L_{\sigma^{-1}(\mathfrak{r})}$ of extensions of $K_{\mathfrak{p}}$. This implies that $g(\sigma(\alpha)) = 0$ in $L_{\mathfrak{r}}$ if and only if $\sigma^{-1}(g(\sigma(\alpha))) = g(\alpha) = 0$ in $L_{\sigma^{-1}(\mathfrak{r})}$. Since $L_{\sigma^{-1}(\mathfrak{r})}$ contains $E_{\sigma^{-1}(\mathfrak{r}) \cap B}$ as a subfield (in fact, the subfield $K_{\mathfrak{p}}(\alpha)$), this completes the proof. $\square$

As a simple example, consider $E = \mathbb{Q}(i)$, $K = \mathbb{Q}$. We have seen that $2\mathcal{O}_E = (1+i)^2$; thus $X^2 + 1$ is irreducible in $\mathbb{Q}_2[X]$. On the other hand, $5\mathcal{O}_E = (2+i)(2-i)$, where $(2+i)$, $(2-i)$ are distinct prime ideals of $\mathcal{O}_E$. It follows that $X^2 + 1$ must factorize in $\mathbb{Q}_5[X]$.

We now describe this factorization. By Hensel's lemma, there is a unique element $\theta \in \mathbb{Z}_5$ such that $\theta^2 = -1$ and $\theta \equiv 2 \pmod{5}$. Then $X^2 + 1 = (X + \theta)(X - \theta)$ in $\mathbb{Q}_5[X]$. Which factor corresponds to which prime ideal of $\mathcal{O}_E$? The ideal $(2 + i)$ corresponds to the factor $g(X)$ such that $g(i) = 0$ in $E_{(2+i)}$. Since $i \equiv -2 \pmod{2+i}$, we see that the corresponding irreducible factor is $X + \theta$. Similarly, the ideal $(2 - i)$ corresponds to the factor $(X - \theta)$.

For another example, we consider the field $E = \mathbb{Q}(\sqrt[3]{2})$, $K = \mathbb{Q}$, and factorise $5\mathcal{O}_E$. The polynomial $X^3 - 2$ is irreducible in $\mathbb{Q}[X]$. Since $3^3 \equiv 2 \pmod{5}$, Hensel's lemma shows that there is a unique $\theta \in \mathbb{Z}_5$ such that $\theta^3 = 2$, and also that there are no other roots of $X^3 - 2$ in $\mathbb{Z}_5$ (since there are none in $\mathbb{F}_5$).

We conclude that $X^3 - 2$ factors as $(X - \theta)g(X)$ in $\mathbb{Q}_5[X]$, where $g(X)$ is irreducible, and hence that there are exactly two distinct prime ideals $\mathfrak{q}_1, \mathfrak{q}_2$ of $\mathcal{O}_E$ lying above $5\mathcal{O}_E$. If $\mathfrak{q}_1$ corresponds to $(X - \theta)$ then $E_{\mathfrak{q}_1} = \mathbb{Q}_5$, while $E_{\mathfrak{q}_2}$ is a quadratic extension of $\mathbb{Q}_5$.

4 Extensions of complete DVRs

We have shown that understanding the decomposition of prime ideals, and their decomposition groups, in an extension of Dedekind domains is equivalent to understanding the Galois theory of extensions of complete fields. Accordingly, we now focus on this case.

Definition 4.1. We call a complete discrete valuation field (CDVF) a pair (K, v_K) consisting of a field K and a valuation $v_K : K^\times \rightarrow \mathbb{Z}$ such that the corresponding DVR $A_K = \{x \in K^\times \mid v_K(x) \geq 0\} \cup \{0\}$ is complete.

We usually say that K is a CDVF, and leave v_K implicit. We write A_K for its discrete valuation ring, $\pi_K \in A_K$ for a choice of uniformizer, and $k_K = A_K/(\pi_K)$ for the residue field.

Lemma 4.2. Let K be a CDVF, and let E/K be a finite separable extension. Then E has a natural structure of CDVF.

Proof. Let B denote the integral closure of A_K in E . It is a Dedekind domain; we first show that B has a unique non-zero prime ideal (as then it is a DVR, which defines the valuation on E). Let $\alpha \in E$ be a primitive element, and let $f(X) \in K[X]$ be its minimal polynomial. By Corollary 3.14, the non-zero prime ideals of B correspond to factors of $f(X)$ in $K[X]$ (as A_K is already complete); so there is a unique such.

We next show that B is complete. Let $\mathfrak{p} \subset A$, $\mathfrak{q} \subset B$ be the non-zero prime ideals, so $\mathfrak{p}B = \mathfrak{q}^{e_{\mathfrak{q}/\mathfrak{p}}}B$. The ring B is finite free as an A -module, so there is an isomorphism $B \cong \varprojlim_i B/\mathfrak{p}^i B = \varprojlim_i B/\mathfrak{q}^{ie_{\mathfrak{q}/\mathfrak{p}}}B$. In other words, B is complete. $\square$

In particular, it makes sense to define $e_{E/K} = e_{\mathfrak{q}/\mathfrak{p}}$ and $f_{E/K} = f_{\mathfrak{q}/\mathfrak{p}}$. Note that by definition, we have $v_E|_{K^\times} = e_{E/K}v_K$.

Definition 4.3. Let E, K be CDVFs. We call an extension E/K an extension of CDVFs if it is finite and separable and v_E is the valuation associated to v_K by Lemma 4.2.

We say that an extension E/K of CDVFs is unramified if $e_{E/K} = 1$, $f_{E/K} = [E : K]$, and the extension k_E/k_K is separable. We say that it is totally ramified if $e_{E/K} = [E : K]$ and $f_{E/K} = 1$.

Corollary 4.4. *Suppose that E/K is a Galois extension of CDVFs. Then for any $x \in E$ and $\sigma \in \text{Gal}(E/K)$, we have $v_E(\sigma(x)) = v_E(x)$.*

Proof. This is true since any $\sigma \in \text{Gal}(E/K)$ determines an automorphism of A_E which acts as the identity on the subring $A_K \subset A_E$. $\square$

Corollary 4.5. *Let E/K be an extension of CDVFs. For any $x \in E^\times$, we have $v_E(x) = \frac{1}{f_{E/K}} v_K(N_{E/K}(x))$.*

Proof. Let L/K be the Galois closure of L . Then for any $x \in E^\times$ we have $N_{L/K}(x) = \prod_{\sigma \in \text{Gal}(L/K)} \sigma(x) = N_{E/K}(x)^{[L:E]}$ and hence

$$\begin{aligned} v_K(N_{E/K}(x)) &= e_{L/K}^{-1} [L : E]^{-1} v_L(N_{L/K}(x)) = e_{L/K}^{-1} [L : E]^{-1} [L : K] v_L(x) \\ &= e_{L/K}^{-1} e_{L/E} [L : E]^{-1} [L : K] v_E(x) = e_{E/K}^{-1} [E : K] v_E(x). \end{aligned}$$

Using the formula $e_{E/K} f_{E/K} = [E : K]$ gives the desired result. $\square$

Working in a complete DVR is very pleasant. We now give several examples of this.

Definition 4.6. *Let A be a DVR and let $v : K^\times \rightarrow \mathbb{Z}$ be the corresponding valuation. If $f(X) = X^d + a_1 X^{d-1} + \cdots + a_d \in K[X]$ is a polynomial with $a_d \neq 0$, then we define the Newton polygon of f to be the graph of the largest continuous piecewise linear function $N : [0, d] \rightarrow \mathbb{R}$ satisfying the following conditions:*

- $N(0) = 0$ and $N(d) = v(a_d)$.
- For all $j = 1, \dots, d$, $N(j) \leq v(a_j)$ if $a_j \neq 0$.
- The graph of N is convex (i.e. the sequence of slopes of line segments of $N_K(f)$ is increasing).

In other words, the Newton polygon is the lower convex hull of the set of points $(j, v(a_j))$.

Proposition 4.7. *Suppose that $f(X)$ factors as $f(X) = \prod_{i=1}^d (X - \alpha_i)$ with $\alpha_i \in K$. Suppose that the Newton polygon of $f(X)$ has slopes $\gamma_1 \leq \gamma_2 \leq \cdots \leq \gamma_d$, counted with multiplicity. Then after re-ordering we have $\gamma_i = v(\alpha_i)$ for each i .*

Proof. Let $\lambda_i = v(\alpha_i)$; we can assume after re-ordering that $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_d$. Define the λ -polygon to be the graph of the continuous piecewise linear function $L : [0, d] \rightarrow \mathbb{R}$ with $L(0) = 0$ and $L'(x) = \lambda_i$ for $x \in (i-1, i)$. Then $L(d) = \lambda_1 + \cdots + \lambda_d = v(\alpha_1 \cdots \alpha_d) = v(a_d)$, so the Newton polygon and the λ -polygon have the same endpoints. We must show that they are in fact the same.

We first show that the Newton polygon lies above the λ -polygon. For this, it suffices to show that for all $j = 1, \dots, d$, we have $v(a_j) \geq \lambda_1 + \dots + \lambda_j$. But we can write $a_j = \pm \sum \alpha_{i_1} \dots \alpha_{i_j}$, and each term in the sum has valuation $\lambda_{i_1} + \dots + \lambda_{i_j} \geq \lambda_1 + \dots + \lambda_j$. We therefore have $v(a_j) \geq \lambda_1 + \dots + \lambda_j$, by the ultrametric property.

We now show that the two polygons are in fact equal. We will use the following general fact: if $x_1, \dots, x_r \in K$ and $v(x_1) < v(x_j)$ for all $j > 1$, then $v(x_1 + \dots + x_r) = v(x_1)$. Suppose that $\lambda_1 = \dots = \lambda_{k_1} < \lambda_{k_1+1} = \dots = \lambda_{k_2} < \dots$. We can write

$$a_{k_j} = \alpha_1 \dots \alpha_{k_j} + \sum \alpha_{i_1} \dots \alpha_{i_{k_j}}.$$

The terms in the sum all involve α_r for some $r > k_j$, hence have valuation strictly greater than $\lambda_1 + \dots + \lambda_{k_j}$. We conclude that $v(a_{k_j}) = v(\alpha_1 \dots \alpha_{k_j}) = \lambda_1 + \dots + \lambda_{k_j}$.

This implies that the two polygons must in fact coincide. Indeed, we have shown that the Newton polygon lies above the λ -polygon, and that the Newton polygon shares a vertex with each of the vertices of the λ -polygon. This completes the proof. $\square$

Corollary 4.8. *Let A be a complete DVR with $\text{Frac}(A) = K$, and let $f(X) = X^d + a_1 X^{d-1} + \dots + a_d \in K[X]$ be a separable polynomial with $a_d \neq 0$. Suppose that the slopes of the Newton polygon of $f(X)$ are $\lambda_1 \leq \dots \leq \lambda_k$, each appearing with width $w_1, \dots, w_k$. Then there is a unique factorization $f(X) = \prod_{i=1}^k g_i(X)$ in $K[X]$ such that $\deg g_i(X) = w_i$ and the Newton polygon of $g_i(X)$ has a single segment of slope λ_i .*

Proof. Let L/K denote the splitting field of $f(X)$. Then L/K is Galois (this is where we use the hypothesis that $f(X)$ is separable – we could alternatively assume that K is perfect, allowing the possibility that $f(X)$ has repeated roots). In particular, L/K is an extension of CDVFs. Let $\alpha_1, \dots, \alpha_n$ be the roots of $f(X)$ in L . We know that the numbers λ_i are exactly the $e_{L/K}^{-1} v_L(\alpha_j)$, with the width w_i being the number of j with $v_L(\alpha_j) = e_{L/K} \lambda_i$. We therefore define $g_i(X) = \prod (X - \alpha_j)$, the product being over the roots with $v_L(\alpha_j) = e_{L/K} \lambda_i$.

We clearly have $f(X) = \prod_{i=1}^k g_i(X)$ in $L[X]$. We claim that each $g_i(X)$ in fact lies in $K[X]$. The group $\text{Gal}(L/K)$ permutes the α_i leaving the values $v_L(\alpha_i)$ invariant, so fixes the coefficients of each $g_i(X)$, so we have $g_i(X) \in K[X]$ by Galois theory. The uniqueness of the given factorization is clear. $\square$

Example 4.9. The polynomial $X^3 + X^2 - 2X + 8$ has 3 distinct roots in $\mathbb{Q}_2$, because its Newton polygon has 3 distinct slopes.

Definition 4.10. *Let A be a discrete valuation ring with field of fractions K , and let $v_K : K^\times \rightarrow \mathbb{Z}$ be the associated valuation. Let $f(X) = X^n + a_1 X^{n-1} + \dots + a_n \in A[X]$ be a polynomial with $a_n \neq 0$. We say that $f(X)$ is Eisenstein if $a_1, \dots, a_{n-1}$ lie in the maximal ideal of A and $v_K(a_n) = 1$.*

Equivalently, a monic polynomial $f(X) \in A[X]$ of degree n and with $f(0) \neq 0$ is Eisenstein if $N_K(f)$ consists of a single line segment of slope $1/n$.

Proposition 4.11. *Let K be a CDVF.*

1. Let $f(X) = X^n + a_1X^{n-1} + \cdots + a_n \in A_K[X]$ be a separable polynomial which is Eisenstein. Then $f(X)$ is irreducible and the extension $E = K[X]/(f(X))$ is totally ramified, and $X \bmod f(X) \in A_E$ is a uniformizer.
2. Suppose conversely that E/K is a totally ramified extension of CDVFs, and let $\pi_E \in E$ be a uniformizer with minimal polynomial $f(X) = X^n + a_1X^{n-1} + \cdots + a_n \in A_K[X]$. Then $n = [E : K]$, $E = K(\pi_E)$, and $f(X)$ is Eisenstein.

Proof. Let L denote the splitting field of $f(X)$, and let $\pi \in L$ be a root, $E = K(\pi)$. Then $N_E(f)$ is the dilation (in the y -direction) of $N_K(f)$ by a factor of $e_{E/K}$. Hence it is a single line segment of slope $e_{E/K}/n$. Since $f(X)$ has a root in E , this slope must be an integer, so $e_{E/K} \geq n$. On the other hand we have $e_{E/K} \leq [E : K] \leq n$. Therefore equality holds, $f(X)$ is irreducible, and E/K is totally ramified. Moreover, $v_E(\pi) = 1$.

Suppose instead that E/K is a totally ramified extension, and let $f(X) \in A_K[X]$ be the minimal polynomial of a uniformizer $\pi_E \in A_E$. Then $N_K(f)$ has a single slope (since all of the roots of $f(X)$ in a splitting field are conjugate over K , so have the same valuation), which equals $e_{E/K}^{-1}$. The y -coordinate of the end-point of $N_K(f)$ is therefore $n/e_{E/K} = v_K(a_n)$. Since $n \leq [E : K] = e_{E/K}$ and $v_K(a_n)$ is an integer, we find $n = [E : K]$ and so $f(X)$ is irreducible and $f(X)$ is Eisenstein. $\square$

Having given a characterization of totally ramified extensions, we turn to the unramified case.

Proposition 4.12. *Let K be a CDVF, and let l/k_K be a finite separable extension. Then there exists an extension L/K of CDVFs and an isomorphism $\iota : l \rightarrow k_L$ with the following properties:*

1. *The extension L/K is unramified.*
2. *For any finite separable extension E/K equipped with a k -embedding $j : l \rightarrow k_E$, there is a unique K -embedding $J : L \rightarrow E$ which induces $j \circ \iota^{-1}$ on residue fields.*

Proof. Let $\bar{\alpha} \in l$ be a primitive element, let $\bar{f}(x) \in k_K[X]$ be its minimal polynomial, and let $f(X) \in A_K[X]$ be a monic polynomial with reduction modulo π_K equal to $\bar{f}(X)$. Then $f(X)$ is irreducible and separable. Indeed, any factorization $f(X) = g(X)h(X)$ in $A_K[X]$ could be reduced modulo π_K to obtain a non-trivial factorization of $\bar{f}(X)$, contradicting the irreducibility of $\bar{f}(X)$. It is separable since the reduction modulo π_K of the discriminant $\text{disc } f$ equals $\text{disc } \bar{f}$ which is a non-zero element of k_K .

Let $L = K[X]/(f(X))$, and let $\alpha = X \bmod (f(X)) \in L$. Then L/K is an extension of CDVFs and $\alpha \in A_L$. We observe that there is a ring homomorphism $A_K[X]/(f(X)) \rightarrow A_L$, hence a ring homomorphism $A_K[X]/(f(X), \pi_K) \cong l \rightarrow A_L/(\pi_K) \rightarrow A_L/(\pi_L) \cong k_L$. Since any homomorphism of fields is injective, this is actually a k_K -embedding $l \rightarrow k_L$. Since $f_{L/K} \leq [L : K] = \deg f = \deg \bar{f} = [l : k_K]$, the map $l \rightarrow k_L$ is an isomorphism, $f_{L/K} = [L : K]$, and the extension L/K is unramified.

We next address the universal property of L/K . Let E/K be an extension of CDVFs. We know that K -embeddings $L \rightarrow E$ are in bijective correspondence with roots of $f(X)$ in E , hence in A_E (since any such root is integral over A_K).

On the other hand, k_K -embeddings $k_L \rightarrow k_E$ are in bijective correspondence with roots of $\bar{f}(X)$ in k_E . Thus we can associate to any $J : L \rightarrow E$ a $j : k_L \rightarrow k_E$ by taking reduction modulo (π_E) of the associated root of $f(X)$.

To complete the proof, we need only observe that the simple version of Hensel's lemma implies that reduction modulo (π_E) gives a bijective correspondence between roots of $f(X)$ in A_E and roots of $\bar{f}(X) \in k_E$; note that since $\bar{f}(X)$ is separable, each root in k_E is necessarily a simple root. $\square$

We now present some important consequences of this result. The first is that for any prime p and any $n \geq 1$, there is a unique unramified extension $E_n/\mathbb{Q}_p$ of degree n , up to isomorphism. Indeed, we know from Galois Theory that there is a unique extension $\mathbb{F}_{p^n}/\mathbb{F}_p$ of degree n , up to isomorphism. The proposition provides an unramified extension $E_n/\mathbb{Q}_p$ equipped with an isomorphism $\iota : \mathbb{F}_{p^n} \rightarrow k_{E_n}$. If $L_n/\mathbb{Q}_p$ is another unramified extension of degree n , then there is an isomorphism $j : \mathbb{F}_{p^n} \rightarrow k_{L_n}$, which is automatically upgraded to an isomorphism $J : E_n \rightarrow L_n$.

The second is that for any extension E/K of CDVFs with separable residue field extension k_E/k_K , there is a unique subextension $E/E_0/K$ with the property that E_0/K is unramified and $k_E = k_{E_0}$. Indeed, the proposition provides an extension E_0/K with an isomorphism $\iota : k_E \rightarrow k_{E_0}$. The universal property then implies the existence of an embedding $J : E_0 \rightarrow E$ which induces ι^{-1} at the level of residue fields, so we can identify E_0 with a subfield of E . We see that E_0 contains any intermediate field $E/E_1/K$ such that E_1/K is unramified; we can therefore call E_0 the maximal unramified subextension of E/K .

When E/K is further Galois, we can characterize E_0 in a different way. Indeed, in this case we have proved that k_E/k_K is Galois and that there is a surjective homomorphism $\text{Gal}(E/K) \rightarrow \text{Gal}(k_E/k_K)$. We define the inertia group $I_{E/K}$ to be the kernel of this homomorphism. Galois Theory implies that $E_1 = E^{I_{E/K}}$ is a subextension of E/K with $\text{Gal}(E_1/K) = \text{Gal}(k_E/k_K)$. We claim that in fact $E_0 = E_1$, and E_1 is the maximal unramified subextension of E/K :

Proposition 4.13. *Let E/K be a Galois extension of CDVFs, with k_E/k_K separable. Then the maximal unramified subextension E_0 of E/K equals $E^{I_{E/K}}$.*

Proof. Let $\bar{\alpha} \in k_E$ be a primitive element, let $\bar{f}(X) \in k_K[X]$ be its minimal polynomial, and let $f(X) \in A_K[X]$ be any monic lift of $\bar{f}(X)$. We have seen that E_0 may be characterized as the subfield of E generated by a root of $f(X)$. On the other hand, Hensel's lemma shows that $f(X)$ splits into linear factors in $A_{E_0}[X]$. Consequently E_0 is the splitting field of $f(X)$, and is a Galois extension of K . The map reduction modulo π_K gives a bijection between the roots of $f(X)$ in E_0 and the roots of $\bar{f}(X)$ in k_E . In particular, an element $\sigma \in \text{Gal}(E/K)$ acts as the identity on E_0 if and only if it acts as the identity on k_E . This shows that $\text{Gal}(E/E_0) = I_{E/K}$, and hence that $E^{I_{E/K}} = E_0$, as claimed. $\square$

Let E/K be a Galois extension of CDVFs such that k_E/k_K is separable.

Definition 4.14. *Let $G = \text{Gal}(E/K)$. For each $i \geq 0$ we define the i^{th} lower ramification group $G_i = \ker(\text{Gal}(E/K) \rightarrow \text{Aut}(A_E/(\pi_E^{i+1})))$. By convention, we take $G_{-1} = G$.*

Thus $G_0 = I_{E/K} = \ker(\text{Gal}(E/K) \rightarrow \text{Gal}(k_E/k_K))$ is the usual inertia group. We have $G_{-1} \supset G_0 \supset G_1 \supset \dots$ and $\cap_{i \geq 0} G_i = \{1\}$. Moreover, each G_i is a normal subgroup of G (it is the kernel of a homomorphism). If $E/L/K$ is an intermediate field extension and $H = \text{Gal}(E/L)$, then it follows from the definition that $H_i = H \cap G_i$. The following lemma is useful for computations.

Lemma 4.15. *Let $\sigma \in G_0$. Then for each $i \geq 0$, $\sigma \in G_i$ if and only if $v_E(\sigma(\pi_E) - \pi_E) \geq i+1$.*

Proof. Let E_0/K denote the maximal unramified subfield of E/K , so $G_0 = \text{Gal}(E/E_0)$. We have $A_E = A_{E_0}[\pi_E]$. Indeed, by Nakayama's lemma this holds provided the map $A_{E_0}[\pi_E] \rightarrow A_E/(\pi_{E_0}) = A_E/(\pi_E^{e_{E/K}})$ is surjective. However, any element of A_E admits a π_E -adic expansion $\sum_{j=0}^{\infty} a_j \pi_E^j$, where the elements a_j lie in A_{E_0} (as the map $A_{E_0} \rightarrow k_E$ is surjective), and any finite such expression lies in $A_{E_0}[\pi_E]$.

We see in particular that $\sigma \in G_0$ acts trivially on $A_E/(\pi_E^{i+1})$ if and only if $\sigma(\pi_E) \equiv \pi_E \pmod{(\pi_E^{i+1})}$, of equivalently $v_E(\sigma(\pi_E) - \pi_E) \geq i+1$. $\square$

Example 4.16. Let $E = \mathbb{Q}_2(\sqrt{2})$, $K = \mathbb{Q}_2$. Then E is the splitting field of the Eisenstein polynomial $X^2 + 2$, so E/K is ramified and we can choose $\pi_E = \sqrt{2}$ as a uniformizer. The Galois group $G = \{1, s\}$ contains two elements, and $v_E(s(\pi_E) - \pi_E) = v_E(-\sqrt{2} - \sqrt{2}) = 3$. Thus $G = G_0 = G_1 = G_2$, and $G_3 = G_4 = \dots$ are trivial.

Example 4.17. Let $E = \mathbb{Q}_2(i)$, $K = \mathbb{Q}_2$. The identity $(1+i)^2 = 2i$ shows that E/K is a ramified quadratic extension and that we can choose $\pi_E = 1+i$ as a uniformizer. The Galois group $G = \{1, t\}$ has two elements, and $v_E(t(\pi_E) - \pi_E) = v_E(-i - i) = v_E(-2i) = 2$. Thus $G = G_0 = G_1$ and $G_2 = G_3 = \dots$ are trivial.

We see that in the first case, we have to go one step deeper into the filtration of G by its lower ramification groups before they become trivial. This is a sense in which the extension $\mathbb{Q}_2(\sqrt{2})$ is “more ramified” than the extension $\mathbb{Q}_2(i)$.

Example 4.18. Let $E = \mathbb{Q}_2(\sqrt{2}, i)$, $K = \mathbb{Q}_2$. Then E is compositum of two ramified quadratic extensions, which are distinct (since they have different ramification groups). Therefore E/K has degree 4 and $\text{Gal}(E/K)$ has 4 elements $\{1, s, t, st\}$, where s sends $\sqrt{2}$ to $-\sqrt{2}$ and fixes i , while t sends i to $-i$ and fixes $\sqrt{2}$.

The element $\zeta = (1+i)/\sqrt{2}$ is a primitive 8th root of unity, and $\zeta - 1$ is a root of the polynomial $(X+1)^4 + 1 = X^4 + 4X^3 + 6X^2 + 4X + 2$, which is Eisenstein over $\mathbb{Q}_2$. We find that E/K is totally ramified, so $G = G_0$, and $\pi_E = \zeta - 1$ is a uniformizer.

We calculate $v_E(\sigma(\pi_E) - \pi_E) = v_E(\sigma(\zeta) - \zeta)$ for the various elements $\sigma \in G$:

$$s : v_E\left(-\frac{1+i}{\sqrt{2}} - \frac{1+i}{\sqrt{2}}\right) = v_E(-2\zeta) = 4,$$

$$t : v_E\left(\frac{1-i}{\sqrt{2}} - \frac{1+i}{\sqrt{2}}\right) = v_E\left(-\frac{2i}{\sqrt{2}}\right) = 2,$$

$$st : v_E\left(-\frac{1-i}{\sqrt{2}} - \frac{1+i}{\sqrt{2}}\right) = v_E(-\sqrt{2}) = 2.$$

We find $G_0 = G_1 = \{1, s, t, st\}$, $G_2 = G_3 = \{1, s\}$, and $G_4 = \{1\}$.

Lemma 4.19. *Let $\pi \in A_L$ be a uniformizer, and let $s \in G_0$, $i \geq 0$. Then $s \in G_i$ if and only if $s(\pi)/\pi \equiv 1 \pmod{(\pi^i)}$.*

Proof. We can assume, after replacing K by the maximal unramified subextension of L/K , that L/K is totally ramified and $A_L = A_K[\pi]$. If $s \in G_0$, then we have $i_G(s) = v_L(s(\pi) - \pi) = v_L(s(\pi)/\pi - 1) + 1$, hence $i_G(s) \geq i + 1$ if and only if $s(\pi)/\pi \equiv 1 \pmod{\pi^i B}$. $\square$

If $s \in G_0$, then the element $s(\pi)/\pi$ lies in $A_L^\times$ (since it has valuation 0). We define a filtration of the group $U_L = A_L^\times$ by the formula $U_L^i = \ker(A_L^\times \rightarrow (A_L/\pi^i)^\times)$. We thus have $U_L/U_L^1 \cong (A_L/\mathfrak{m}_L)^\times = k_L^\times$. On the other hand for $i \geq 1$ we have $U_L^i = 1 + \pi^i A_L$, and we have an isomorphism $\pi^i A_L/\pi^{i+1} A_L \cong U_L^i/U_L^{i+1}$ given by $x \mapsto 1 + x$. It is a homomorphism because $(1+x)(1+y) = 1 + x + y + xy$, and $xy \in \pi^{i+1} A_L$.

Proposition 4.20. *Let $\pi \in A_L$ be a uniformizer.*

1. *There is an injection $G_0/G_1 \rightarrow k_L^\times$ given by the formula $s \mapsto s(\pi)/\pi \pmod{\mathfrak{m}_L}$.*
2. *For each $i \geq 1$, there is an injection $G_i/G_{i+1} \rightarrow \pi^i A_L/\pi^{i+1} A_L$ given by the formula $s \mapsto s(\pi)/\pi - 1$.*
3. *The quotient G_0/G_1 is cyclic. If k_L has characteristic 0, then the group G_1 is trivial. If k_L has characteristic $p > 0$, then the group G_1 is the unique p -Sylow subgroup of G_0 .*

Proof. We have already proved the first two parts. We prove the third. The group G_0 is finite, and G_0/G_1 is a finite subgroup of the multiplicative group of a field, which is therefore cyclic. If k_L has characteristic 0, then each of the quotients G_i/G_{i+1} , $i > 0$, is a finite group which injects into a $\mathbb{Q}$ -vector space, which is therefore trivial. Since $\cap_i G_i = \{1\}$, we see that the group G_1 must be trivial.

If k_L has characteristic p , then the same argument shows that G_1 has order a power of p : each G_i/G_{i+1} is a subgroup of the additive group of a 1-dimensional k_L -vector space. On the other hand, the quotient G_0/G_1 injects into the torsion subgroup of $k_L^\times$, which has order prime to p . The result follows. $\square$

Corollary 4.21. *The group $I_{L/K} = G_0$ is soluble. If the residue field k_K is finite, then the group $\text{Gal}(L/K)$ is soluble. There is no Galois extension $E/\mathbb{Q}_p$ with Galois group A_5 .*

Definition 4.22. *If L/K is a Galois extension of complete discrete non-archimedean valued fields, we say that L/K is tamely ramified if the group G_1 is trivial. We say that it is wildly ramified if G_1 is non-trivial.*

Corollary 4.23. *Suppose that the extension L/K is Galois and totally tamely ramified of degree n . Then K contains the n^{th} roots of unity, and there exists a uniformizer $\pi_K \in A_K$ such that $L = K(\sqrt[n]{\pi_K})$.*

Proof. There is an embedding $G_0 \hookrightarrow k_K^\times$. Any finite subgroup of $k_K^\times$ is cyclic, so we find that G_0 is cyclic of degree n . By the existence of Teichmüller representatives, we find that K itself contains the n^{th} roots of unity. Moreover, if $\pi_L \in A_L$ is a uniformizer and $\sigma \in \text{Gal}(L/K)$ is a generator, then there is a primitive n^{th} root of unity $\zeta \in A_K^\times$ such that $\sigma(\pi_L) = \zeta \pi_L \pmod{\pi_L^2}$.

Let $\alpha = \pi_L + \zeta^{-1}\sigma(\pi_L) + \zeta^{-2}\sigma^2(\pi_L) + \cdots + \zeta^{-(n-1)}\sigma^{n-1}(\pi_L)$. Then $\sigma(\alpha) = \sigma(\pi_L) + \cdots + \zeta\pi_L = \zeta\alpha$, hence $\alpha^n \in A_K$. We claim that α is a uniformizer of A_L . We calculate modulo π_L^2 : here, we have

$$\alpha \bmod \pi_L^2 \equiv \pi_L + \zeta^{-1}\zeta\pi_L + \zeta^{-2}\zeta^2\pi_L + \cdots = n\pi_L \bmod \pi_L^2.$$

Since L/K is tamely ramified, $n \in A_K^\times$ and hence $v(\alpha) = v(\pi_L) = 1$. It follows that $A_L = A_K[\alpha]$ and $\alpha^n = \pi_K$ is a uniformizer of A_K . $\square$

We now continue to investigate the situation where E/K is a Galois extension of CDVFs with k_E/k_K separable. Consider an intermediate extension $E/L/K$ with L/K Galois, and let $H = \text{Gal}(E/L)$, a normal subgroup of G . It is immediate from the definition that lower ramification subgroups are compatible with passage to subgroups: $H_i = G_i \cap H$ for each $i \geq 0$. However, this is not true for passage to quotient, as the following example shows:

Example 4.24. Let $E = \mathbb{Q}_2(\sqrt{2}, i)$. We have seen that $G = \text{Gal}(E/\mathbb{Q}_2) \cong (\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$. We have $G = G_0 = G_1$, and $G_2 = G_3 = \text{Gal}(E/\mathbb{Q}_2(i))$, $G_4 = \{1\}$.

If $L = \mathbb{Q}_2(\sqrt{2})$, $H = \text{Gal}(E/L)$, $G/N = \{1, t\}$, then $(G/H)_0 = (G/H)_1 = (G/H)_2$, $(G/H)_3 = \{1\}$. In particular, $(G/H)_3$ is not equal to the image of G_3 in G/H .

We now show how to modify the numbering of ramification subgroups so that they become compatible with passage to quotient. This work will later play an essential role in our formulation of global class field theory.

Definition 4.25. If $u \geq 0$ is a real number, then we define $G_u = G_{[u]}$. The ramification function is $\varphi_{E/K} : [0, \infty) \rightarrow [0, \infty)$ is defined by $\varphi_{E/K}(u) = \int_{t=0}^u [G_0 : G_t]^{-1} dt$.

We observe that φ is a continuous, piecewise linear, strictly increasing homeomorphism of $[0, \infty)$ to itself. The points of discontinuity of its derivative occur at integer values of u .

Definition 4.26. For $v \in [0, \infty)$, we define $\psi_{E/K}(v) = \varphi_{E/K}^{-1}(v)$. Thus $\psi_{E/K}$ is a continuous piecewise linear, increasing homeomorphism of $[0, \infty)$ to itself. The ramification groups G^v are then defined for all real numbers $v \in [0, \infty)$ by the formula $G^v = G_{\psi_{E/K}(v)}$.

It is clear from the definition that $\cap_{v>0} G^v = \{1\}$ and that $G^0 = G_0 = I_{E/K}$. The filtration is left-continuous, in the sense that $G^v = \cap_{\epsilon>0} G^{v-\epsilon}$ for all $v \geq 0$. We say that a real number v is a jump in the upper-numbering filtration if the inclusion $G^{v+\epsilon} \subset G^v$ is strict for all $\epsilon > 0$. In contrast to the lower numbering, jumps can occur at rational numbers which are not integers! (See Question 4 on the example sheet for an example of this behaviour.)

Example 4.27. Let $E = \mathbb{Q}_2(\sqrt{2}, i)$, $L = \mathbb{Q}_2(\sqrt{2})$. Let $G = \text{Gal}(E/\mathbb{Q}_2)$, $H = \text{Gal}(E/L)$. We calculate $G^1 = G$, $G^2 = \text{Gal}(E/\mathbb{Q}_2(i))$ (and the jumps are at $v = 1, 2$). We have $(G/H)^2 = G^2H/H$.

We now give another description of the function φ . We first introduce an important function $i_G : G \rightarrow \mathbb{Z}_{\geq 0} \rightarrow \infty$ by the formula $i_G(s) = 1 + \sup\{i \geq -1 \mid s \in G_i\}$ (and $i_G(1) = \infty$, by convention). Thus we have the formulae

$$G_u = \{s \in G \mid i_G(s) \geq u + 1\}$$

and for any $s, t \in G$,

$$i_G(st) \geq \min(i_G(s), i_G(t)),$$

with equality if $i_G(s) \neq i_G(t)$.

Lemma 4.28. *Suppose there exists $\alpha \in A_E$ such that $A_E = A_K[\alpha]$. Then for any $s \in G$, $i_G(s) = v_E(s(\alpha) - \alpha)$.*

Proof. Let $i \in \mathbb{N}$. Then $i_G(s) \geq i + 1$ if and only if $s \in G_i$, if and only if s acts trivially on $A_E/(\pi_E^{i+1})$, if and only if s acts trivially on the image of α in $A_E/(\pi_E^{i+1})$, if and only if $s(\alpha) \equiv \alpha(\pi_E^{i+1})$, if and only if $v_E(s(\alpha) - \alpha) \geq i + 1$. $\square$

Lemma 4.29. *There exists $\alpha \in A_E$ such that $A_E = A_K[\alpha]$.*

Proof. Let $\bar{y} \in k_E$ be a primitive element for k_E/k_K , let $\bar{f}(X) \in k_K[X]$ be its minimal polynomial, let $f(X) \in A_K[X]$ be a monic lift of $\bar{f}(X)$, and let $y \in A_E$ be the unique root of $f(X)$ which is congruent to $\bar{y}$ modulo (π_E) (this exists by Hensel's lemma). Let $\alpha = y + \pi_E$. We claim that $A_E = A_K[\alpha]$.

By Nakayama's lemma, it is enough to show that the map $A_K[\alpha] \rightarrow A_E/(\pi_E) = A_E/(\pi_E^{e_{E/K}})$ is surjective. We note that $A_K[\alpha]$ contains a set of representatives for the residue field k_E (as $\alpha \bmod (\pi_E) = \bar{y}$). Moreover, we have $f(\alpha) = f(y + \pi_E) = f'(y)\pi_E + \pi_E^2 z$ for some $z \in A_E$. Since $\bar{f}(X)$ is separable, $f'(y)$ is a unit and so $f(\alpha)$ has valuation 1, hence is a uniformizer. Therefore any element of $A_E/(\pi_E^{e_{E/K}})$ admits a representative of the form $\sum_{i=0}^{e_{E/K}-1} a_i f(\alpha)^i$, with $a_i \in A_K[\alpha]$. Since also $f(\alpha) \in A_K[\alpha]$, this completes the proof. $\square$

Lemma 4.30. *We have the formula $\varphi_{E/K}(u) + 1 = \frac{1}{\#G_0} \sum_{s \in G} \min(i_G(s), u + 1)$.*

Proof. Both functions are continuous and piecewise linear, and take the value 1 at $u = 0$. It therefore suffices to show that for non-integer values of u , the left-hand side and the right-hand side have the same derivative.

At a point $m < u < m + 1$ with $m \in \mathbb{N}$, the left-hand side has derivative $[G_0 : G_u]^{-1}$. The function $\min(i_G(s), u + 1)$ has derivative 0 if $i_G(s) < u + 1$ and derivative 1 if $i_G(s) > u + 1$ (equivalently, if $s \in G_u$). Thus the derivative of the right-hand side is $|G_0|^{-1} |G_u| = [G_0 : G_u]^{-1}$, as desired. $\square$

Lemma 4.31. *Let $s \in G$, let H be a normal subgroup of G , and let $L = E^H$. Then we have the formula*

$$i_{G/H}(sH) = \frac{1}{e_{E/L}} \sum_{t \in H} i_G(st).$$

Proof. We can assume that $s \notin H$ (otherwise both sides are ∞). Choose elements $\alpha \in A_E$, $\beta \in A_L$ such that $A_E = A_K[\alpha]$ and $A_L = A_K[\beta]$. Then we have

$$i_{G/H}(sH) = v_L(s(\beta) - \beta) = \frac{1}{e_{E/L}} v_E(s(\beta) - \beta),$$

while

$$\frac{1}{e_{E/L}} \sum_{t \in H} i_G(st) = \frac{1}{e_{E/L}} v_E\left(\prod_{t \in H} (st(\alpha) - \alpha)\right).$$

It is therefore enough to show that $s(\beta) - \beta$ and $\prod_{t \in H}(st(\alpha) - \alpha)$ have the same valuation (under v_E). To this end, let $f(X) = \sum_i a_i X^i \in A_L[X]$ denote the minimal polynomial of α over L , so $f(X) = \prod_{t \in H}(X - t(\alpha))$. Let $s(f)(X) = \sum_i s(a_i)X^i = \prod_{t \in H}(X - st(\alpha)) \in A_E[X]$. Then we have

$$\prod_{t \in H}(st(\alpha) - \alpha) = s(f)(\alpha) - f(\alpha) = \sum_i (s(a_i) - a_i)\alpha^i.$$

Each element $s(a_i) - a_i$ is divisible by $s(\beta) - \beta$, so we see that $v_E(\prod_{t \in H}(st(\alpha) - \alpha)) \geq v_E(s(\beta) - \beta)$.

For the reverse inequality, choose $g(X) \in A_K[X]$ such that $g(\alpha) = \beta$. Then $g(X) - \beta \in A_L[X]$ has α as a root, so is divisible by $f(X)$ in $A_L[X]$. It follows that $g(X) - s(\beta)$ is divisible by $s(f)(X)$ in $A_E[X]$, and so $g(\alpha) - s(\beta) = \beta - s(\beta)$ is divisible by $s(f)(\alpha) = \prod_{t \in H}(st(\alpha) - \alpha)$ and $v_E(\prod_{t \in H}(st(\alpha) - \alpha)) \leq v_E(s(\beta) - \beta)$. This completes the proof. $\square$

Lemma 4.32. *Let H be a normal subgroup of G , and let $L = E^H$. For $s \in G$, define $j(sH) = \sup\{i_G(st) \mid t \in H\}$. Then $i_{G/H}(sH) = 1 + \varphi_{E/L}(j(sH) - 1)$.*

Proof. We have just shown that $i_{G/H}(sH) = \frac{1}{e_{E/L}} \sum_{t \in H} i_G(st)$. On the other hand, we have

$$1 + \varphi_{E/L}(j(sH) - 1) = \frac{1}{|H_0|} \sum_{t \in H} \min(i_H(t), j(sH)).$$

Noting that $|H_0| = e_{E/L}$ and $i_H(t) = i_G(t)$ for $t \in H$, we see that it is enough to show that $i_G(st) = \min(i_H(t), j(sH))$ for each $t \in H$. We will show that this is true provided that the representative s for the coset sH is chosen so that $i_G(s) = j(sH)$ (i.e. so that $i_G(st) \leq i_G(s)$ for all $t \in H$).

In this case, we have $\min(i_H(t), j(sH)) = \min(i_G(t), i_G(s))$. If $i_G(s) \neq i_G(t)$ then $i_G(st) = \min(i_G(t), i_G(s))$ (as observed above), so both sides agree. If $i_G(s) = i_G(t)$, then $i_G(st) \geq i_G(s)$, hence $i_G(st) = i_G(s)$ (as $i_G(s)$ is maximal), so again both sides agree. This concludes the proof. $\square$

Theorem 4.33 (Herbrand's Theorem). *Let H be a normal subgroup of G , and let $L = E^H$. Let $u \in [0, \infty)$, and let $v = \varphi_{E/L}(u)$. Then $(G/H)_v = G_u H/H$.*

Proof. For $s \in G$, we have $sH \in G_u H/H$ if and only if $j(sH) \geq u + 1$, if and only if $j(sH) - 1 \geq u$, if and only if $\varphi_{E/L}(j(sH) - 1) \geq \varphi_{E/L}(u) = v$, if and only if $i_{G/H}(sH) \geq v + 1$, if and only if $sH \in (G/H)_v$. $\square$

Lemma 4.34. *We have the formula $\varphi_{E/K} = \varphi_{L/K} \circ \varphi_{E/L}$.*

Proof. Let H be a normal subgroup of G , and let $L = E^H$. Both sides are continuous, piecewise linear, increasing homeomorphisms of $[0, \infty)$ into itself. It suffices therefore to show that the derivatives are equal, wherever they are defined. The derivative of the left-hand side at u equals $[G_0 : G_u]^{-1}$, while the derivative of the right-hand side equals $\varphi'_{L/K}(\varphi_{E/L}(u))\varphi'_{E/L}(u) = [(G/H)_0 : (G/H)_{\varphi_{E/L}(u)}]^{-1}[H_0 : H_u]^{-1}$. By the previous lemma, this is equal to $[(G/H)_0 : G_u H/H]^{-1}[H_0 : H_u]^{-1}$. Noting that $(G/H)_0 = G_0 H/H$, we see that $[(G/H)_0 : G_u H/H] = |G_0||H_u||H_0|^{-1}|G_u|^{-1}$, which gives the desired result. $\square$

Theorem 4.35. *Let H be a normal subgroup of G , and let $L = E^H$. For all $v \geq 0$, we have $(G/H)^v = G^v H/H$.*

Proof. By the lemma, we have $\psi_{E/K} = \psi_{E/L} \circ \psi_{L/K}$. By definition, we have $(G/H)^v = (G/H)_{\psi_{L/K}(v)}$. By Herbrand's theorem, we have $(G/H)_{\psi_{L/K}(v)} = G_{\psi_{E/L} \circ \psi_{L/K}(v)} H/H = G_{\psi_{E/K}(v)} H/H = G^v H/H$. $\square$

The existence of the upper numbering allows us to give a robust generalization of the notion of maximal unramified subextension.

Definition 4.36. *Let E/K be a (possibly not Galois) extension of CDVFs with k_E/k_K separable, and let $v \geq 0$ be a real number. We define $E^v = E \cap L^{G^v}$, where L/E is extension of CDVFs with k_L/k_E separable such that L/K is Galois and $G = \text{Gal}(L/K)$.*

Thus if $v < v'$ then $E^v \subset E^{v'}$, and $\cup_{v \geq 0} E^v = E$. We check that this definition is independent of the choice of L . It suffices to check that if L'/L is a further extension of CDVFs with $k_{L'}/k_L$ separable and L'/K Galois then $E \cap L^{G^v} = E \cap (L')^{(G')^v}$ (where $G' = \text{Gal}(L'/K)$). This follows from the equality $(G')^v \text{Gal}(L'/L) / \text{Gal}(L'/L) = G^v$.

Theorem 4.37. *Let E/K be a (possibly not Galois) extension of CDVFs with k_E/k_K separable.*

1. E^0 is the maximal unramified subextension of E/K .
2. Let $E/M/K$ be an intermediate subfield. Then $M^v = E^v \cap M$.
3. Let $E/M, N/K$ be intermediate subfields. Then $M^v \cdot N^v \subset (M \cdot N)^v$. In particular, if $M^v = M$ and $N^v = N$ then $(M \cdot N)^v = M \cdot N$.

Proof. Let L/K be the Galois closure of E/K , and let $G = \text{Gal}(L/K)$. We have $G^v = G^0 = G_0 = I_{L/K}$, so the first part is clear. The second part follows from the definition. For the third, the given inclusion may be rewritten as $(M \cap L^{G^v}) \cdot (N \cap L^{G^v}) \subset (M \cdot N) \cap L^{G^v}$, which is again clear. $\square$

This notion becomes particularly useful in the case of abelian extensions of p -adic fields, thanks to the following theorem, which will not be proved in this course:

Theorem 4.38 (Hasse–Arf Theorem). *Let $K/\mathbb{Q}_p$ be a finite extension, and let E/K be an abelian extension (i.e. a finite Galois extension with abelian Galois group). Let $v \geq 0$ be a jump in the ramification filtration (i.e. a real number such that $\text{Gal}(E/K)^v \neq \text{Gal}(E/K)^{v+\epsilon}$ for all $\epsilon > 0$). Then $v \in \mathbb{Z}$.*

This theorem in turn motivates the following definition, which we will use to organize our formulation of global class field theory.

Definition 4.39. *Let $K/\mathbb{Q}_p$ be a finite extension, and let E/K be an abelian extension. We define the conductor ideal of E/K to be the ideal $C_{E/K} = (\pi_K^a)$, where $a \geq 0$ is the smallest non-negative integer such that $\text{Gal}(E/K)^a = \{1\}$ (or equivalently $E^a = E$).*

Note that $C_{E/K}$ is the unit ideal if and only if E/K is unramified.

Proposition 4.40. *Let $K/\mathbb{Q}_p$ be a finite extension, and let E/K be a Galois extension. Let $E_1, E_2/K$ be subextensions of E/K which are abelian over K . Then $E_1 \cdot E_2$ is abelian over K and $C_{E_1 \cdot E_2/K} = \text{lcm}(C_{E_1/K}, C_{E_2/K})$.*

Proof. There is an injective homomorphism $\text{Gal}(E_1 \cdot E_2/K) \rightarrow \text{Gal}(E_1/K) \times \text{Gal}(E_2/K)$ (given by restricting a given homomorphism to each subfield). Since a product of abelian groups is abelian, $E_1 \cdot E_2/K$ is an abelian extension.

If $a \in [0, \infty)$, then we have $(E_1 \cdot E_2)^a = (E_1 \cdot E_2) \cap E^a$, so $E_1 \cdot E_2 = (E_1 \cdot E_2)^a$ if and only if $E_1^a = E_1$ and $E_2^a = E_2$. This shows $C_{E_1 \cdot E_2/K} = \text{lcm}(C_{E_1/K}, C_{E_2/K})$. $\square$

5 Global class field theory

We now begin our discussion of global class field theory. Class field theory aims to describe all abelian extensions of a given number field K . By definition, an abelian extension is a finite Galois extension E/K with abelian Galois group $\text{Gal}(E/K)$. We begin by describing the shape of the theory in the case $K = \mathbb{Q}$.

Let $N \geq 1$ be an integer, and let $\zeta_N = e^{2\pi i/N}$. Then the extension $\mathbb{Q}(\zeta_N)/\mathbb{Q}$ is abelian: it is Galois, because it is the splitting field of the polynomial $X^N - 1$, and it is abelian because there is an isomorphism $\text{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q}) \rightarrow (\mathbb{Z}/N\mathbb{Z})^\times$ which sends $\sigma \in \text{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q})$ to the unique residue class $a \bmod N$ such that $\sigma(\zeta_N) = \zeta_N^a$. We write $\sigma_a \in \text{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q})$ for the automorphism corresponding to $a \bmod N$.

By Galois theory, the following two sets are in canonical bijection:

- The set of abelian extensions $K/\mathbb{Q}$ such that $K \subset \mathbb{Q}(\zeta_N)$.
- The set of subgroups $H \subset (\mathbb{Z}/N\mathbb{Z})^\times$.

A field K corresponds to the image of $\text{Gal}(\mathbb{Q}(\zeta_N)/K)$ in $(\mathbb{Z}/N\mathbb{Z})^\times$. To see how this describes all abelian extensions, we need the Kronecker–Weber theorem:

Theorem 5.1. *Let $K/\mathbb{Q}$ be an abelian extension. Then there exists an integer $N \geq 1$ such that $K \subset \mathbb{Q}(\zeta_N)$. More precisely, we have $K \subset \mathbb{Q}(\zeta_N)$ if and only if $C_{K/\mathbb{Q}}|(N)$ (divisibility of ideals of $\mathbb{Z}$).*

This allows us to upgrade our correspondence to a bijection between the following two sets:

- The set of abelian extensions $K/\mathbb{Q}$ such that $C_{K/\mathbb{Q}}|(N)$.
- The set of subgroups of $H \subset (\mathbb{Z}/N\mathbb{Z})^\times$.

This is close to what class field theory will say when $\mathbb{Q}$ is replaced by a general number field. However, the definition of the above bijection still depends on our explicit description of $\mathbb{Q}(\zeta_N)$ as an abelian extension with Galois group $(\mathbb{Z}/N\mathbb{Z})^\times$. In general, class field theory

will describe the Galois groups of abelian extensions explicitly, but it will not describe the extensions themselves. We therefore need to give a characterization of our bijection that does not involve the knowledge of explicit abelian extensions of $\mathbb{Q}$.

This is based on the existence of Frobenius elements. Recall that if K is a number field and E/K is a Galois extension (not necessarily abelian), then the Frobenius element $\text{Frob}_{Q/P} \in G = \text{Gal}(E/K)$ is defined for any non-zero prime ideal $P \subset \mathcal{O}_K$ lying below the prime ideal $Q \subset \mathcal{O}_E$ such that $e_{Q/P} = 1$. It may be characterized as the unique element of the decomposition group $D_{Q/P}$ which induces the Frobenius automorphism $x \mapsto x^{|k_P|}$ on the residue field k_Q .

If $\sigma \in G$, then $\sigma(Q)$ is another prime ideal of $\mathcal{O}_E$ lying above P , so $\text{Frob}_{\sigma(Q)/P} \in G$ is also defined. In fact, $\text{Frob}_{\sigma(Q)/P} = \sigma \text{Frob}_{Q/P} \sigma^{-1}$. To see this, note that $\sigma \text{Frob}_{Q/P} \sigma^{-1} \in D_{\sigma(Q)/P}$, and σ descends to an isomorphism $k_Q \rightarrow k_{\sigma(Q)}$ of extensions of k_P which intertwines the Frobenius automorphism on source and target. In particular, when G is abelian, $\text{Frob}_{\sigma(Q)/P} = \text{Frob}_{Q/P}$ is independent of the choice of prime ideal of $\mathcal{O}_E$ lying above P . We record this fact as a definition:

Definition 5.2. *Let E/K be an abelian extension of number fields, and let $P \subset \mathcal{O}_K$ be a non-zero prime ideal which is unramified in $\mathcal{O}_E$. Then the Artin symbol $(P, E/K) \in \text{Gal}(E/K)$ is defined to be the Frobenius element $\text{Frob}_{Q/P}$ for some (hence any) choice of prime ideal $Q \subset \mathcal{O}_E$ lying above P .*

We can now complete our formulation of class field theory for cyclotomic fields.

Theorem 5.3. *Let $N \geq 1$ be an integer, and let $K/\mathbb{Q}$ be an abelian extension such that $C_{K/\mathbb{Q}} | N$. In particular, any prime $p \nmid N$ is unramified, so the Artin symbol $((p), K/\mathbb{Q}) \in \text{Gal}(K/\mathbb{Q})$ is defined. Then there is a unique surjective homomorphism $\phi_{K/\mathbb{Q}} : (\mathbb{Z}/N\mathbb{Z})^\times \rightarrow \text{Gal}(K/\mathbb{Q})$ which, for any prime $p \nmid N$, sends the residue class $p \bmod N$ to the Artin symbol $((p), K/\mathbb{Q})$. The assignment $K \mapsto \ker \phi_{K/\mathbb{Q}}$ defines a bijection between the following two sets:*

- *The set of abelian extensions $K/\mathbb{Q}$ such that $C_{K/\mathbb{Q}} | (N)$.*
- *The set of subgroups of $H \subset (\mathbb{Z}/N\mathbb{Z})^\times$.*

This is the statement that we will generalize to an arbitrary base number field. Before we do so, let's check that the bijection of the theorem agrees with the previous one. In other words, we must check that the homomorphism $(\mathbb{Z}/N\mathbb{Z})^\times \rightarrow \text{Gal}(K/\mathbb{Q})$ arising from an embedding $K \subset \mathbb{Q}(\zeta_N)$ agrees with $\phi_{K/\mathbb{Q}}$; or in other words, that if $p \nmid N$ is a prime then $\sigma_p|_K = ((p), K/\mathbb{Q})$. It follows from the definitions that $((p), K/\mathbb{Q}) = ((p), \mathbb{Q}(\zeta_N)/\mathbb{Q})|_K$, so it's enough to show that $\sigma_p = ((p), \mathbb{Q}(\zeta_N)/\mathbb{Q})$. The automorphism σ_p is characterized by $\sigma_p(\zeta_N) = \zeta_N^p$. On the other hand, if $P \subset \mathcal{O}_{\mathbb{Q}(\zeta_N)}$ is a prime ideal lying above (p) , then $((p), \mathbb{Q}(\zeta_N)/\mathbb{Q})$ is characterized by

$$((p), \mathbb{Q}(\zeta_N)/\mathbb{Q})(\zeta_N \bmod P) = \zeta_N^p \bmod P.$$

Since the map $\mathcal{O}_{\mathbb{Q}(\zeta_N)} \rightarrow k_P$ induces a bijection on N^{th} roots of unity, we must in fact have

$$((p), \mathbb{Q}(\zeta_N)/\mathbb{Q})(\zeta_N) = \zeta_N^p,$$

hence $\sigma_p = ((p), \mathbb{Q}(\zeta_N)/\mathbb{Q})$.

We now turn to the general case. We first need to introduce a certain amount of notation. Let K be a number field.

Definition 5.4. A modulus of K is a pair $\mathfrak{m} = (\mathfrak{m}_0, \mathfrak{m}_\infty)$, where $\mathfrak{m}_0 \subset \mathcal{O}_K$ is a non-zero ideal and $\mathfrak{c}_\infty \subset \text{Hom}_{\mathbb{Q}}(K, \mathbb{R})$ is a subset (which may be empty).

If $\mathfrak{m}, \mathfrak{n}$ are moduli we write $\mathfrak{m} \leq \mathfrak{n}$ if $\mathfrak{m}_0 | \mathfrak{n}_0$ and $\mathfrak{m}_\infty \subset \mathfrak{n}_\infty$. This defines a partial order on the set of moduli of K .

The trivial modulus is $(\mathcal{O}_K, \emptyset)$. It satisfies $(\mathcal{O}_K, \emptyset) \leq \mathfrak{m}$ for any other modulus $\mathfrak{m}$. If E/K is any abelian extension, then we can define a modulus $\mathfrak{m}_{E/K} = (\mathfrak{m}_{E/K,0}, \mathfrak{m}_{E/K,\infty})$, where $\mathfrak{m}_{E/K,0} = C_{E/K}$ is the conductor ideal and $\mathfrak{m}_{E/K,\infty}$ is the set of embeddings $\tau : K \rightarrow \mathbb{R}$ such that there does not exist an embedding $\tilde{\tau} : E \rightarrow \mathbb{R}$ with $\tilde{\tau}|_K = \tau$ (more informally, the set of real embeddings of K which do not extend to real embeddings of E).

Definition 5.5. Let $\mathcal{I}$ denote the group of non-zero fractional ideals of $\mathcal{O}_K$, and $\mathcal{P} \subset \mathcal{I}$ its subgroup of non-zero principal fractional ideals (i.e. of the form $\alpha \mathcal{O}_K$ for some $\alpha \in K^\times$).

If $\mathfrak{m} = (\mathfrak{m}_0, \mathfrak{m}_\infty)$ is a modulus, then we write $\mathcal{I}(\mathfrak{m}_0) \subset \mathcal{I}$ for the subgroup of non-zero fractional ideals I which are prime to $\mathfrak{m}_0$, in the sense that for any non-zero prime ideal $P \subset \mathcal{O}_K$ such that $v_P(\mathfrak{m}_0) > 0$, we have $v_P(I) = 0$. We write $K(\mathfrak{m}_0) \subset K^\times$ for the subgroup of elements $\alpha \in K^\times$ such that for any non-zero prime ideal $P \subset \mathcal{O}_K$ such that $v_P(\mathfrak{m}_0) > 0$, we have $v_P(\alpha) = 0$. We write $K_{\mathfrak{m}} \subset K(\mathfrak{m}_0)$ for the subgroup of elements $\alpha \in K^\times$ such that for any non-zero prime ideal $P \subset \mathcal{O}_K$ such that $v_P(\mathfrak{m}_0) > 0$, we have $v_P(\alpha - 1) \geq v_P(\mathfrak{m}_0)$, and for any $\tau \in \mathfrak{m}_\infty$, we have $\tau(\alpha) > 0$.

We write $\mathcal{P}(\mathfrak{m}_0) \subset \mathcal{P}$ for the subgroup of principal non-zero fractional ideals of the form (α) for some $\alpha \in K(\mathfrak{m}_0)$. We write $\mathcal{P}_{\mathfrak{m}} \subset \mathcal{P}(\mathfrak{m}_0)$ for the subgroup of principal fractional ideals of the form (α) for some $\alpha \in K_{\mathfrak{m}}$.

Finally, we define the ray class group of modulus $\mathfrak{m}$ to be the quotient $H(\mathfrak{m}) = \mathcal{I}(\mathfrak{m}_0)/\mathcal{P}_{\mathfrak{m}}$.

We observe that the ray class group associated to the trivial modulus is the quotient $H_K = \mathcal{I}/\mathcal{P}$; this is the usual ideal class group of the number field K . In general, the ray class group generalizes simultaneously the usual ideal class group of the number field K and the group $(\mathbb{Z}/N\mathbb{Z})^\times$ in the case $K = \mathbb{Q}$. This is made precise by the following proposition:

Proposition 5.6. Let $\mathfrak{m}$ be a modulus of K .

1. The ray class group $H(\mathfrak{m})$ is a finite abelian group.
2. There are short exact sequences of abelian groups

$$0 \longrightarrow \mathcal{P}(\mathfrak{m}_0)/\mathcal{P}_{\mathfrak{m}} \longrightarrow H(\mathfrak{m}) \longrightarrow H_K \longrightarrow 0$$

$$0 \longrightarrow \mathcal{O}_K^\times / (\mathcal{O}_K^\times \cap K_{\mathfrak{m}}) \longrightarrow (\mathcal{O}_K/\mathfrak{m}_0)^\times \times \prod_{\tau \in \mathfrak{m}_\infty} \{\pm 1\} \longrightarrow \mathcal{P}(\mathfrak{m}_0)/\mathcal{P}_{\mathfrak{m}} \longrightarrow 0.$$

Proof. We take on trust the theorem, proved in Part II Number Fields, that the ideal class group H_K is always finite. The first part of the proposition then follows from the second, using the fact that given a short exact sequence

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

of abelian groups, the finiteness of M is equivalent to the finiteness of M' and M'' (and in this case we have $|M| = |M'| \cdot |M''|$).

The first exact sequence we write down is

$$0 \longrightarrow \mathcal{I}(\mathfrak{m}_0)/\mathcal{P}(\mathfrak{m}_0) \longrightarrow H(\mathfrak{m}) \longrightarrow \mathcal{I}(\mathfrak{m}_0)/\mathcal{P}(\mathfrak{m}_0) \longrightarrow 0.$$

This will give the first sequence in the statement of the proposition if we can construct an isomorphism $\mathcal{I}(\mathfrak{m}_0)/\mathcal{P}(\mathfrak{m}_0) \rightarrow H_K = \mathcal{I}/\mathcal{P}$. However, $\mathcal{P}(\mathfrak{m}_0) = \mathcal{I}(\mathfrak{m}_0) \cap \mathcal{P}$, so there is a natural injection $\mathcal{I}(\mathfrak{m}_0)/\mathcal{P}(\mathfrak{m}_0) \rightarrow H_K$. To show that it is surjective, we must show that for any non-zero fractional ideal I of $\mathcal{O}_K$, there is $\alpha \in K^\times$ such that $\alpha^{-1}I \in \mathcal{I}(\mathfrak{m}_0)$. We can assume without loss of generality that $I \subset \mathcal{O}_K$ is a non-zero ideal. By the Chinese remainder theorem, we can find $\alpha \in K^\times$ such that $v_{\mathfrak{p}}(\alpha) = v_{\mathfrak{p}}(I)$ if $v_P(\mathfrak{m}_0) > 0$. Then $v_P(\alpha^{-1}I) = 0$ if $v_P(\mathfrak{m}_0) > 0$, hence $\alpha^{-1}I \in \mathcal{I}(\mathfrak{m}_0)$.

To get the second exact sequence, we use the commutative diagram with exact rows:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{O}_K^\times & \longrightarrow & K(\mathfrak{m}_0) & \longrightarrow & \mathcal{P}(\mathfrak{m}_0) \longrightarrow 0 \\ & & \uparrow & & \uparrow & & \uparrow \\ 0 & \longrightarrow & \mathcal{O}_K^\times \cap K_{\mathfrak{m}} & \longrightarrow & K_{\mathfrak{m}} & \longrightarrow & \mathcal{P}_{\mathfrak{m}} \longrightarrow 0. \end{array}$$

By the snake lemma, there is an exact sequence

$$0 \longrightarrow \mathcal{O}_K^\times/(\mathcal{O}_K^\times \cap K_{\mathfrak{m}}) \longrightarrow K(\mathfrak{m}_0)/K_{\mathfrak{m}} \longrightarrow \mathcal{P}(\mathfrak{m}_0)/\mathcal{P}_{\mathfrak{m}} \longrightarrow 0.$$

We need to construct an isomorphism

$$K(\mathfrak{m}_0)/K_{\mathfrak{m}} \rightarrow (\mathcal{O}_K/\mathfrak{m}_0)^\times \times \prod_{\tau \in \mathfrak{m}_\infty} \{\pm 1\}$$

is an isomorphism. We first construct a map. Note that if $P \subset \mathcal{O}_K$ is a non-zero prime ideal such that $v_P(\mathfrak{m}_0) > 0$, then the inclusion $K(\mathfrak{m}_0) \subset K_P^\times$ factors through an inclusion $K(\mathfrak{m}_0) \rightarrow \mathcal{O}_{K_P}^\times$, and so there is a map $K(\mathfrak{m}_0) \rightarrow \mathcal{O}_{K_P}^\times/(1 + \mathfrak{m}_0\mathcal{O}_{K_P}) \cong (\mathcal{O}_K/P^{v_P(\mathfrak{m}_0)})^\times$, hence (by the Chinese remainder theorem) a map $K(\mathfrak{m}_0) \rightarrow (\mathcal{O}_K/\mathfrak{m}_0)^\times$. We take the product of this map with the map $K(\mathfrak{m}_0) \rightarrow \prod_{\tau \in \mathfrak{m}_\infty} \{\pm 1\}$ which sends $\alpha \in K(\mathfrak{m}_0)$ to $(\text{sgn}\tau(\alpha))_{\tau \in \mathfrak{m}_\infty}$. This determines a homomorphism

$$K(\mathfrak{m}_0) \rightarrow (\mathcal{O}_K/\mathfrak{m}_0)^\times \times \prod_{\tau \in \mathfrak{m}_\infty} \{\pm 1\}$$

with kernel $K_{\mathfrak{m}}$. To complete the proof, we just need to show that this homomorphism is surjective. Let $N \in \mathbb{Z} \cap \mathfrak{m}_0$ be a positive integer. By the Chinese remainder theorem, for any

$x \in (\mathcal{O}_K/\mathfrak{m}_0)^\times$, we can find $\alpha \in \mathcal{O}_K$ such that $\alpha \equiv x \pmod{\mathfrak{m}_0}$. After replacing α by $\alpha + kN$ for some $k \in \mathbb{N}$, we can assume moreover that $\tau(\alpha) > 0$ for each $\tau \in \mathfrak{m}_\infty$.

We therefore just need to show that for any subset $S \subset \mathfrak{m}_\infty$, we can find $\beta \in \mathcal{O}_K$ such that $\beta \equiv 1 \pmod{\mathfrak{m}_0}$, $\tau(\beta) > 0$ if $\tau \in S$, and $\tau(\beta) < 0$ if $\tau \notin S$. To accomplish this, choose $\gamma \in \mathcal{O}_K$ such that $K = \mathbb{Q}(\gamma)$, and let $f(X) \in \mathbb{Z}[X]$ be the minimal polynomial of γ over $\mathbb{Q}$. The embeddings of K in $\mathbb{R}$ correspond to the real roots of $f(X)$. Let $\gamma_1, \dots, \gamma_r$ be the roots of $f(X)$ corresponding to the embeddings $\tau_1, \dots, \tau_r$ in $\mathfrak{m}_\infty$. We can find a polynomial $g(X) \in \mathbb{R}[X]$ such that $g(\gamma_i) = 1$ if $\tau_i \in S$ and $g(\gamma_i) = -1$ if $\tau_i \notin S$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$, we can find a polynomial $h(X) \in \mathbb{Q}[X]$ such that $h(\gamma_i) > 0$ if $\tau_i \in S$ and $h(\gamma_i) < 0$ if $\tau_i \notin S$. Clearing denominators, we can assume moreover that $h(X) \in \mathbb{Z}[X]$. The element $\beta = 1 + kNh(\gamma) \in \mathcal{O}_K$ then does the job for any sufficiently large $k \in \mathbb{N}$. $\square$

Example 5.7. Suppose that $K = \mathbb{Q}$ and $\mathfrak{m} = ((N), \tau)$, for some integer $N \geq 1$. Then $\mathbb{Z}^\times = \{\pm 1\}$ and $\mathbb{Z}^\times \cap \mathbb{Q}_\mathfrak{m}$ is trivial. The ideal class group $H_\mathbb{Q}$ is trivial, so the proposition gives an isomorphism $H(\mathfrak{m}) \cong ((\mathbb{Z}/N\mathbb{Z})^\times \times \{\pm 1\})/\{\pm 1\} \cong (\mathbb{Z}/N\mathbb{Z})^\times$ in this case. If $I \subset \mathbb{Z}$ is an ideal which is prime to N , then this isomorphism sends the class $[I]$ of I in $H(\mathfrak{m})$ to the integer m , where $m \geq 1$ is the unique *positive* generator of I .

We now consider an example where the embeddings of K in $\mathbb{R}$ play a role. Let $d > 1$ be a squarefree integer, and let $K = \mathbb{Q}(\sqrt{d})$ be a real quadratic field. Let $\mathfrak{m} = (\mathcal{O}_K, \{\tau_1, \tau_2\})$, where $\tau_1, \tau_2 : K \rightarrow \mathbb{R}$ are the two distinct real embeddings. Suppose further that the ideal class group H_K is trivial. Dirichlet's unit theorem shows that there is a unit $\epsilon \in \mathcal{O}_K^\times$ such that $\mathcal{O}_K^\times = \{\pm \epsilon^n \mid n \in \mathbb{Z}\}$. The proposition shows that $H(\mathfrak{m}) = (\{\pm 1\} \times \{\pm 1\})/\langle \pm \epsilon^\mathbb{Z} \rangle$. Thus $H(\mathfrak{m})$ is trivial if and only if $\tau_1(\epsilon)$ and $\tau_2(\epsilon)$ have opposite signs. Since $\mathbf{N}_{K/\mathbb{Q}}(\epsilon) = \tau_1(\epsilon)\tau_2(\epsilon) \in \mathbb{Z}^\times = \{\pm 1\}$, we find that $H(\mathfrak{m})$ is trivial if $N_{K/\mathbb{Q}}(\epsilon) = -1$ and $H(\mathfrak{m})$ is cyclic of order 2 if $N_{K/\mathbb{Q}}(\epsilon) = 1$.

Either case can occur: examples are given by $\mathbb{Q}(\sqrt{2})$ (which has class number 1 and $\epsilon = 1 + \sqrt{2}$) and $\mathbb{Q}(\sqrt{3})$ (which has class number 1 and $\epsilon = 2 + \sqrt{3}$).

We can now state the main theorem of global class field theory.

Theorem 5.8. *Let K be a number field, and let $\mathfrak{m}$ be a modulus of K .*

1. *Let E/K be an abelian extension such that $\mathfrak{m}_{E/K} \leq \mathfrak{m}$. Then there is a unique surjective homomorphism $\phi_{E/K} : H(\mathfrak{m}) \rightarrow \text{Gal}(E/K)$ which, for any non-zero prime ideal $P \subset \mathcal{O}_K$ such that $v_P(\mathfrak{m}_0) = 0$, sends the class $[P] \in H(\mathfrak{m})$ to the Artin symbol $(P, E/K) \in \text{Gal}(E/K)$.*
2. *The assignment $E \mapsto \ker \phi_{E/K}$ defines an inclusion-reversing bijection between the following two sets:*
 - *The set of abelian extensions E/K such that $\mathfrak{m}_{E/K} \leq \mathfrak{m}$.*
 - *The set of subgroups $H \subset H(\mathfrak{m})$.*

In particular, there exists a maximal abelian extension $L(\mathfrak{m})/K$ such that $\mathfrak{m}_{L(\mathfrak{m})/K} \leq \mathfrak{m}$; for this extension, $\phi_{L(\mathfrak{m})/K}$ determines an isomorphism $H(\mathfrak{m}) \rightarrow \text{Gal}(L(\mathfrak{m})/K)$. We call $L(\mathfrak{m})$ the ray class field of modulus $\mathfrak{m}$.

We now discuss examples. The basic case is when $\mathfrak{m} = (\mathcal{O}_K, \emptyset)$ is the trivial modulus. In this case, class field theory says that there is a maximal abelian extension L/K which is everywhere unramified and in to which every real embedding of K extends; it is called the Hilbert class field, and there is an isomorphism $\phi_{L/K} : H_K \rightarrow \text{Gal}(L/K)$.

Example 5.9. Consider the polynomial $f(X) = X^3 - X + 1$. As $X^3 + aX + b$ has discriminant $-4a^3 - 27b^2$, the polynomial $f(X)$ has discriminant -23 , and is irreducible (even over $\mathbb{F}_3$). Thus the splitting field L of $f(X)$ has Galois group S_3 over $\mathbb{Q}$, and contains the quadratic extension $K = \mathbb{Q}(\sqrt{-23})$. We claim that L is the Hilbert class field of K . For this we need to know two things:

- The extension L/K is everywhere unramified. (Since K has no real embeddings, there is no additional condition.) This will show that L is contained inside the Hilbert class field of K .
- The group H_K has order 3. This will show that L actually equals the Hilbert class field of K .

The class number can be calculated using either the Minkowski bound (as in Part II Number Fields) or using the theory of binary quadratic forms (on which more shortly). To show that L/K is everywhere unramified, we observe that the polynomial $f(X)$ has distinct roots modulo p for every prime $p \neq 23$, so any prime $p \neq 23$ is unramified in L . There is a unique prime ideal $P \subset \mathcal{O}_K$ lying above 23, namely $(\sqrt{-23})$. We need to check that P is unramified in L . Equivalently, we must check that if $Q \subset \mathcal{O}_L$ is a prime ideal lying above 23, then $e_{Q/(23)}$ is prime to 3.

To this end, we observe that $3f(X) - Xf'(X) = -3X + 3 + X = -2X + 3$ is linear. This implies that $\bar{f}(X) = f(X) \bmod 23$ has two distinct roots in $\mathbb{F}_{23}$, one occurring with multiplicity 2. If $\alpha_1, \alpha_2, \alpha_3 \in \mathcal{O}_L$ are the roots of $f(X)$, then $\bar{\alpha}_1 = \alpha_1 \bmod Q$, $\alpha_2 \bmod Q$, $\alpha_3 \bmod Q$ are the roots (with multiplicity) of $\bar{f}(X)$ in k_Q . Therefore we can assume after relabelling that $\bar{\alpha}_1 \neq \bar{\alpha}_2 = \bar{\alpha}_3$. An element of the inertia group $I_{Q/(23)}$ acts as the identity on the residue field, so must fix α_1 and possibly permute α_2 and α_3 . Since $I_{Q/(23)}$ is non-trivial (because 23 is ramified in $\mathcal{O}_L$) we see that $I_{Q/(23)}$ is generated by a transposition, and indeed has order prime to 3.

Example 5.10. For an example where the real embeddings play a role, let us consider the field $K = \mathbb{Q}(\sqrt{3})$. We have already observed that H_K is trivial, but that if $\mathfrak{m} = (\mathcal{O}_K, \{\tau_1, \tau_2\})$ then $H(\mathfrak{m})$ is cyclic of order 2. Class field theory tells us that K has no everywhere unramified extension to which the real embeddings extend, but there does admit an everywhere unramified quadratic extension E/K which admits no real embeddings. What is it? It is given by $E = K(i) = K(\sqrt{-3})$. (These two presentations of E illustrate that E/K is everywhere unramified, since $\mathbb{Q}(i)$ is ramified only at 2 and $\mathbb{Q}(\sqrt{-3})$ is ramified only at 3.)

In the remainder of the course, we explain the relation between the class field theory of imaginary quadratic fields and binary quadratic forms.

Definition 5.11. A binary quadratic form is a function $f(x, y) = ax^2 + bxy + cy^2$ where a, b, c are integers. We say that the form f represents a given integer m if there exist values $x_0, y_0 \in \mathbb{Z}$ such that $f(x_0, y_0) = m$.

We can equivalently represent a binary quadratic form as

$$f(x, y) = (x, y) \begin{pmatrix} a & b/2 \\ b/2 & c \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

We will analyse the following problem: when is a given prime p represented by a given positive definite binary quadratic form $f(x, y)$? Here are some classical results on this problem: Let $p > 7$ be a prime. Then $p = x^2 + y^2$ if and only if $p \equiv 1 \pmod{4}$ (Fermat). We also have $p = x^2 + 2y^2$ if and only if $p \equiv 1, 3 \pmod{8}$ (Euler), and $p = x^2 + 5y^2$ if and only if $p \equiv 1, 9 \pmod{20}$ (Gauss).

On the other hand, one can show that $p = x^2 + 14y^2$ if and only if the equations $x^2 = -14$ and $(y^2 + 1)^2 = 8$ have a solution in $\mathbb{F}_p$. In general, if $n \equiv 1, 2$ is a squarefree positive integer, then a prime $p \nmid 4n$ is represented by the binary quadratic form $x^2 + ny^2$ if and only if p splits completely in the Hilbert class field of the imaginary quadratic field $\mathbb{Q}(\sqrt{-n})$.

To explain this we first need to understand the explicit relation between ideal class groups of imaginary quadratic fields and binary quadratic forms. This is organized around the notion of discriminant.

Definition 5.12. Let $f(x, y) = ax^2 + bxy + cy^2$ be a binary quadratic form. Its discriminant is $\text{disc } f = b^2 - 4ac$.

We observe that a binary quadratic form is positive definite (i.e. takes only positive values on non-zero vectors) if and only if $\text{disc } f < 0$ and $a > 0$.

The group $\text{SL}_2(\mathbb{Z})$ acts on the set of binary quadratic forms by the formula

$$(\gamma \cdot f)(x, y) = f(\gamma^{-1} \begin{pmatrix} x \\ y \end{pmatrix}).$$

Equivalently, if f is represented by the matrix $\begin{pmatrix} a & b/2 \\ b/2 & c \end{pmatrix}$, then $\gamma \cdot f$ is represented by the matrix $\gamma^{-t} \begin{pmatrix} a & b/2 \\ b/2 & c \end{pmatrix} \gamma^{-1}$. Since the determinant of the matrix equals $-\frac{1}{4} \text{disc } f$, this shows that $\text{disc } f$ is invariant under the action of $\text{SL}_2(\mathbb{Z})$.

We also need to recall the notion of discriminant in number fields. If K is a number field of degree $[K : \mathbb{Q}] = n$, and $\alpha_1, \dots, \alpha_n \in K$, then we define the discriminant $\text{disc}(\alpha_1, \dots, \alpha_n) = \det(D)^2$, where $D \in M_n(\mathbb{C})$ is the matrix $D_{ij} = \sigma_i(\alpha_j)$, and where $\sigma_1, \dots, \sigma_n : K \rightarrow \mathbb{C}$ are the distinct embeddings of K in the complex numbers. The quantity $\text{disc}(\alpha_1, \dots, \alpha_n)$ is independent of the choice of ordering of $\sigma_1, \dots, \sigma_n$ because any permutation changes $\det(D)$ by a sign, which disappears when we take the square. For similar reasons it is independent of the choice of ordering of $\alpha_1, \dots, \alpha_n$.

More generally, if $A \in M_n(\mathbb{Z})$ and we define $\beta_1, \dots, \beta_n \in K$ by $\beta_i = \sum_{j=1}^n A_{ij} \alpha_j$, then $\text{disc}(\beta_1, \dots, \beta_n) = \det(A)^2 \text{disc}(\alpha_1, \dots, \alpha_n)$. This formula is particularly useful when we are given a subgroup $M \subset K$ which is free of rank n as a $\mathbb{Z}$ -module; then we can define $\text{disc } M = \text{disc}(\alpha_1, \dots, \alpha_n)$, where $\alpha_1, \dots, \alpha_n$ is any $\mathbb{Z}$ -basis of M . Since any two bases are

related by an element of $\mathrm{GL}_n(\mathbb{Z})$, our formula shows that this is independent of the choice of basis.

If $M' \subset M \subset K$ are two subgroups which are free of rank n as $\mathbb{Z}$ -modules, then a very similar argument (using Smith normal form) shows that $\mathrm{disc} M' = \mathrm{disc} M \cdot [M : M']^2$. In particular, if $\mathrm{disc} M' = \mathrm{disc} M$ then $M' = M$.

A key example is $M = \mathcal{O}_K$. Indeed, $\mathcal{O}_K$ is always free of rank n as a $\mathbb{Z}$ -module, so its discriminant $\mathrm{disc} \mathcal{O}_K$ is defined. We compute this invariant when K is a quadratic field. Fix therefore $d \in \mathbb{Z}$ a squarefree integer, $d \neq 0, 1$. We have seen that if $d \equiv 2, 3 \pmod{4}$, then $\mathcal{O}_K = \mathbb{Z} \oplus \mathbb{Z}\sqrt{d}$, so we have

$$\mathrm{disc} \mathcal{O}_K = \det \begin{pmatrix} 1 & \sqrt{d} \\ 1 & -\sqrt{-d} \end{pmatrix}^2 = 4d.$$

The other case is when $d \equiv 1 \pmod{4}$, in which case $\mathcal{O}_K = \mathbb{Z} \oplus \mathbb{Z}\frac{1+\sqrt{d}}{2}$, so we have

$$\mathrm{disc} \mathcal{O}_K = \det \begin{pmatrix} 1 & \frac{1+\sqrt{d}}{2} \\ 1 & \frac{1-\sqrt{d}}{2} \end{pmatrix}^2 = d.$$

We can now formulate the fundamental relation between ideal class groups of imaginary quadratic fields and binary quadratic forms:

Theorem 5.13. *Let $d < 0$ be a squarefree integer, and let $K = \mathbb{Q}(\sqrt{d})$, where $\sqrt{d} \in \mathbb{C}$ is the square root with positive imaginary part. Let $D = \mathrm{disc} \mathcal{O}_K$. Then there is a bijection between the following two sets:*

- *The set of $\mathrm{SL}_2(\mathbb{Z})$ -orbits of positive definite binary quadratic forms of discriminant D .*
- *The ideal class group H_K .*

This bijection sends the orbit of the binary quadratic form $f(x, y) = ax^2 + bxy + cy^2$ to the class of the fractional ideal $\mathbb{Z} \oplus \mathbb{Z}\beta$ of $\mathcal{O}_K$, where $\beta = (-b + \sqrt{D})/2a$ (and $\sqrt{D}$ has positive imaginary part).

We first prove the following intermediate result:

Proposition 5.14. *With the same assumptions, there is a bijection between the following two sets:*

- *The set of positive definite binary quadratic forms of discriminant D .*
- *The set of elements $\beta \in K$ of positive imaginary part such that $\mathbb{Z} \oplus \mathbb{Z}\beta$ is a fractional ideal of $\mathcal{O}_K$.*

This bijection sends the binary quadratic form $f(x, y) = ax^2 + bxy + cy^2$ to the element $\beta = (-b + \sqrt{D})/2a$. It is equivariant for the action of $\mathrm{SL}_2(\mathbb{Z})$, when $\mathrm{SL}_2(\mathbb{Z})$ acts on β by Möbius transformations $\begin{pmatrix} X & Y \\ Z & W \end{pmatrix} \cdot \beta = (X\beta + Y)/(Z\beta + W)$.

Proof. Let $f(x, y) = ax^2 + bxy + cy^2$ be a binary quadratic form of discriminant D and let $\beta = (-b + \sqrt{D})/2a$. We first show that $I = \mathbb{Z} \oplus \mathbb{Z}\beta$ is indeed a fractional ideal of $\mathcal{O}_K$. It is finitely generated as a $\mathbb{Z}$ -module, so we just need to check it is stable under multiplication by elements of $\mathcal{O}_K$. We note that $f(\beta, 1) = 0$ (this is just the quadratic formula), hence $a\beta^2 = -(b\beta + c)$. This shows that $a\beta \cdot I \subset I$, hence $a\beta$ is integral over $\mathbb{Z}$, and $a\beta \in \mathcal{O}_K$, hence $\mathbb{Z} \oplus \mathbb{Z}a\beta \subset \mathcal{O}_K$. We have $\text{disc}(1, a\beta) = a^2(\beta - \bar{\beta})^2 = D = \text{disc } \mathcal{O}_K$, so $\mathbb{Z} \oplus \mathbb{Z}a\beta = \mathcal{O}_K$ and I is indeed a fractional ideal of $\mathcal{O}_K$.

Now suppose $\beta \in K$ has positive imaginary part and $\mathbb{Z} \oplus \mathbb{Z}\beta$ is a fractional ideal of $\mathcal{O}_K$. We associate it to the unique binary quadratic form $f(x, y) = ax^2 + bxy + cy^2$ with the property that $f(\beta, 1) = 0$, $a > 0$, and a, b, c are coprime. We need to check that $f(x, y)$ has discriminant D . To this end, we again note that $a\beta$ leaves $I = \mathbb{Z} \oplus \mathbb{Z}\beta$ stable, so $a\beta \in \mathcal{O}_K$.

If $A + B\beta \in \mathcal{O}_K$ is a general element (with $A, B \in \mathbb{Q}$), then $(A + B\beta)I \subset I$, hence $A, B \in \mathbb{Z}$ and $-Bc/a, A - bB/a \in \mathbb{Z}$, hence $Bc/a \in \mathbb{Z}$ and $bB/a \in \mathbb{Z}$. Since a, b, c are coprime we can find integers x, y, z such that $xa + yb + zc = 1$, hence $x + yb/a + zc/a = 1/a$. This shows that $B/a \in \mathbb{Z}$, hence $B \in a\mathbb{Z}$ and $\mathcal{O}_K = \mathbb{Z} \oplus \mathbb{Z}a\beta$. The discriminant of $(1, a\beta)$ equals $\text{disc } f$ (same computation as before), so we have $\text{disc } f = D$, as required.

We have now constructed two mutually inverse bijections. It remains to explain why they respect the action of $\text{SL}_2(\mathbb{Z})$. Let $\gamma = \begin{pmatrix} X & Y \\ Z & W \end{pmatrix} \in \text{SL}_2(\mathbb{Z})$. Since $f(x, y)$ is homogeneous of degree 2, we have $f(\gamma \cdot \beta) = f((X\beta + Y)/(Z\beta + W), 1) = 0$ if and only if $f(X\beta + Y, Z\beta + W) = f(\gamma \begin{pmatrix} \beta \\ 1 \end{pmatrix}) = 0$. In particular, $(\gamma \cdot f)(\gamma \cdot \beta) = 0$ if and only if $f(\gamma^{-1}\gamma \begin{pmatrix} \beta \\ 1 \end{pmatrix}) = f(\beta) = 0$. □

We can now prove Theorem 5.13:

Proof of Theorem 5.13. Passing to $\text{SL}_2(\mathbb{Z})$ -orbits in the statement of the proposition, it is enough to show that the following two sets are in bijection:

- The set of $\text{SL}_2(\mathbb{Z})$ -orbits of elements $\beta \in K$ of positive imaginary part such that $\mathbb{Z} \oplus \mathbb{Z}\beta$ is a fractional ideal of $\mathcal{O}_K$.
- The set of non-zero fractional ideals of $\mathcal{O}_K$, up to $K^\times$ -multiple.

We first show that if $\beta, \beta' \in K$ lie in the same $\text{SL}_2(\mathbb{Z})$ -orbit, then they define the same class of fractional ideals. Let $\begin{pmatrix} X & Y \\ Z & W \end{pmatrix} \in \text{SL}_2(\mathbb{Z})$, and suppose that $\beta' = (X\beta + Y)/(Z\beta + W)$. Then $\mathbb{Z} \oplus \mathbb{Z}\beta' = \mathbb{Z} \oplus \mathbb{Z}(X\beta + Y)/(Z\beta + W) = (Z\beta + W)^{-1}(\mathbb{Z}(Z\beta + W) \oplus \mathbb{Z}(X\beta + Y))$. However, we have $\mathbb{Z}(Z\beta + W) \oplus \mathbb{Z}(X\beta + Y) = \mathbb{Z} \oplus \mathbb{Z}\beta$ (as subgroups of K), so β and β' do indeed define the same ideal class.

This shows that the map from the first set to the second is well-defined. We next show it is surjective. Any non-zero fractional ideal I of $\mathcal{O}_K$ is free of rank 2 as a $\mathbb{Z}$ -module, so admits a $\mathbb{Z}$ -basis $I = \mathbb{Z}\alpha_1 \oplus \mathbb{Z}\alpha_2$. After re-ordering, we can assume that α_2/α_1 has

positive imaginary part, hence the ideal class $\alpha^{-1}I = \mathbb{Z} \oplus \mathbb{Z}\alpha_2/\alpha_1$ is the image of the orbit of $\beta = \alpha_2/\alpha_1$.

We finally show that our map is injective. Suppose that the ideals $\mathbb{Z} \oplus \mathbb{Z}\beta$ and $\mathbb{Z} \oplus \mathbb{Z}\beta'$ lie in the same class, so there exists $\alpha \in K^\times$ such that $\mathbb{Z} \oplus \mathbb{Z}\beta = \mathbb{Z}\alpha \oplus \mathbb{Z}\beta'\alpha$. Then there exists $\gamma = \begin{pmatrix} X & Y \\ Z & W \end{pmatrix} \in \mathrm{GL}_2(\mathbb{Z})$ such that $\alpha = Z\beta + W$ and $\beta'\alpha = X\beta + Y$, hence $\beta' = (X\beta + Y)/(Z\beta + W)$. Since β, β' both have positive imaginary part, we see that $\det \gamma > 0$, hence $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ and β, β' lie in the same $\mathrm{SL}_2(\mathbb{Z})$ -orbit. $\square$

This theorem allows us to give another proof that the ideal class group H_K of the imaginary quadratic field $K = \mathbb{Q}(\sqrt{d})$ is finite. For this we need to recall the description of the fundamental domain for the action of $\mathrm{SL}_2(\mathbb{Z})$ on the complex upper half plane

$$\mathfrak{h} = \{z \in \mathbb{C} \mid \mathrm{Im} z > 0\}.$$

Consider the set

$$\mathcal{D} = \{z \in \mathfrak{h} \mid |z| > 1, -\frac{1}{2} \leq \mathrm{Re} z < \frac{1}{2}\} \cup \{z \in \mathfrak{h} \mid |z| = 1, -\frac{1}{2} \leq \mathrm{Re} z \leq 0\}.$$

We quote the following proposition without proof.

Proposition 5.15. *Let $z \in \mathfrak{h}$. Then there exists exactly one $z' \in \mathcal{D}$ which is $\mathrm{SL}_2(\mathbb{Z})$ -conjugate to z .*

Corollary 5.16. *Let $D = \mathrm{disc} \mathcal{O}_K$. Then each orbit of positive definite binary quadratic forms of discriminant D contains exactly one form $f(x, y) = ax^2 + bxy + cy^2$ which is reduced, in the sense that $c \geq a \geq |b|$ and if either $c = a$ or $a = |b|$ then $b \geq 0$.*

Proof. The proposition shows that there is precisely one form $f(x, y) = ax^2 + bxy + cy^2$ in each $\mathrm{SL}_2(\mathbb{Z})$ -orbit such that $\beta = (-b + \sqrt{D})/2a$ lies in $\mathcal{D}$. We show that this condition is equivalent to the form $f(x, y)$ being reduced.

We have $|\beta|^2 = (b^2 - (b^2 - 4ac)/4a^2) = c/a$ and $\mathrm{Re} \beta = -b/2a$. Thus the conditions $|\beta| > 1$, $-\frac{1}{2} \leq \mathrm{Re} \beta < \frac{1}{2}$ are equivalent to $c/a > 1$ and $-1/2 \leq -b/2a < 1/2$, or equivalently $c > a$ and $a \geq b > -a$. The conditions $|\beta|^2 = 1$, $-\frac{1}{2} \leq \mathrm{Re} \beta \leq 0$ are equivalent to $c = a$ and $a \geq b \geq 0$. $\square$

Corollary 5.17. *The ideal class group H_K is finite.*

Proof. We need to show that there are finitely many reduced forms of discriminant D . If $f(x, y) = ax^2 + bxy + cy^2$ is reduced then $-D = 4ac - b^2 \geq 3b^2$, so $|b| \leq \sqrt{|D|/3}$. Once b is fixed there are finitely many possibilities for $f(x, y)$. $\square$

The following lemma is useful for doing computations.

Lemma 5.18. *1. The identity element of H_K corresponds to the reduced form $x^2 - Dy^2/4$ (if $D = 4d$) or $x^2 + xy + (1 - D)y^2/4$ (if $D = d$).*

2. Let $f(x, y) = ax^2 + bxy + y^2$ be a reduced form of discriminant D . Then $f(x, y)$ corresponds to an element of order dividing 2 in the ideal class group if and only if either $c = a$ or $a = b$ or $b = 0$.

We call the forms appearing in part 1 of the lemma the principal binary quadratic forms of discriminant D .

Proof. For the first part, we note that the unit ideal $\mathcal{O}_K$ lies in the trivial ideal class, and we have $\mathcal{O}_K = \mathbb{Z} \oplus \mathbb{Z}\sqrt{d}$ or $\mathbb{Z} \oplus \mathbb{Z}(-1 + \sqrt{d})/2$, depending on the value of $d \bmod 4$. In the first case, the binary quadratic form corresponding to $\beta = \sqrt{d}$ is $x^2 - dx^2$, which is visibly reduced. In the second, it is $x^2 + xy + (1 - d)/4$, which is again visibly reduced.

For the second part, we first make the important observation that the Galois group $\text{Gal}(K/\mathbb{Q}) = \{1, \sigma\}$ acts on H_K (here σ denotes complex conjugation). Indeed, it acts on the group $\mathcal{I}$ of non-zero fractional ideals and preserves the subgroup of principal fractional ideals. Moreover, for any ideal class $[I]$ we have $[I][\sigma(I)] = [\mathcal{O}_K]$. Since the classes of non-zero prime ideals generate H_K , it's enough to check this when $I = P$ is a prime ideal of $\mathcal{O}_K$. If P lies above the ideal (p) of $\mathbb{Z}$, then we know that either $p\mathcal{O}_K = P^2$ (if p is ramified), $p\mathcal{O}_K = P$ (if p is inert), or $p\mathcal{O}_K = P\sigma(P)$ (if p splits – recall that $\text{Gal}(K/\mathbb{Q})$ acts transitively on the prime ideals of $\mathcal{O}_K$ lying above (p)). In all cases $P\sigma(P)$ is principal, so $[P][\sigma(P)] = [\mathcal{O}_K]$.

It follows that $[I]$ has order dividing 2 in H_K if and only if $[I] = [\sigma(I)]$. Note that if $I = \mathbb{Z} \oplus \mathbb{Z}\beta$, then $\sigma(I) = \mathbb{Z} \oplus \mathbb{Z}(-\sigma(\beta))$, and $-\sigma(\beta)$ has positive imaginary part. If β corresponds to the positive definite binary quadratic form $ax^2 + bxy + cy^2$, then $-\sigma(\beta)$ corresponds to the form $ax^2 - bxy + cy^2$.

We now split into cases. We assume that the form $f(x, y)$ is reduced, or equivalently that β lies in the fundamental domain $\mathcal{D}$. If β lies in the interior of $\mathcal{D}$, then so does $-\sigma(\beta)$, and so $ax^2 - bxy + cy^2$ is reduced. We thus have $[I] = [\sigma(I)]$ if and only if these two forms are equal, or equivalently if $b = 0$.

Now suppose that β lies in the boundary of $\mathcal{D}$, or equivalently that $c = a$ or $b = a$. In this case $-\sigma(\beta)$ lies on the boundary of the closure of $\mathcal{D}$, but not in $\mathcal{D}$ itself (unless $\beta = i$, in which case $f(x, y)$ is the principal form of discriminant -4 – we can exclude this case from our considerations). The form $ax^2 - bxy + cy^2$ is therefore not reduced, but we see that there is an element of $\text{SL}_2(\mathbb{Z})$ which conjugates $-\sigma(\beta)$ to β (for example, if β satisfies $\text{Re}\beta = -1/2$, then $\beta = -\sigma(\beta) - 1$ is the image of $-\sigma(\beta)$ under the matrix $\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$). Therefore the ideal class $[I]$ has order dividing 2 in this case. $\square$

Corollary 5.19. *Let p be a prime number not dividing D . Then principal form of discriminant D represents p if and only if p splits completely in the Hilbert class field L/K of K .*

Proof. By global class field theory, there is an isomorphism $\phi_{L/K} : H_K \rightarrow \text{Gal}(L/K)$ which sends the class $[P]$ of any non-zero prime ideal $P \subset \mathcal{O}_K$ to the Artin symbol $(P, L/K) \in \text{Gal}(L/K)$.

The prime p splits completely in the extension $L/\mathbb{Q}$ if and only if p splits completely $p = P_1 P_2$ in $K/\mathbb{Q}$ and each prime P_1, P_2 splits completely in L/K . Note that P_1 splits completely if and only if the decomposition group at P_1 inside $\text{Gal}(L/K)$ is trivial. Since it is generated by the Artin symbol $(P_1, L/K)$, this is equivalent to the Artin symbol being trivial. By class field theory, this is equivalent to the prime ideal P_1 being principal.

Thus p splits completely in the extension $L/\mathbb{Q}$ if and only if p splits completely $p = P_1 P_2$ in $K/\mathbb{Q}$ and each prime ideal P_1, P_2 is principal. We must show that if $p \nmid D$ then this condition is equivalent to p being represented by the principal form $f(x, y)$.

However, we can write $f(x, y) = (x + y\beta)(x + y\sigma(\beta))$, where $\mathcal{O}_K = \mathbb{Z} \oplus \mathbb{Z}\beta$. Therefore $f(x, y)$ represents the prime number p if and only if there exists $\alpha \in \mathcal{O}_K$ such that $(\alpha)(\sigma(\alpha)) = p\mathcal{O}_K$. If $p \nmid D$ this is equivalent to the condition appearing in the previous paragraph. $\square$

More generally, one can show that if $f(x, y)$ is a positive definite binary quadratic form of discriminant $D = \text{disc } \mathcal{O}_K$, corresponding to the ideal class $[I_f] \in H_K$, then a given prime number $p \nmid D$ is represented by $f(x, y)$ if and only if p splits completely $p\mathcal{O}_K = P_1 P_2$ and one of P_1, P_2 lies in the ideal class $[I_f]$.

We now return to our numerical examples. If $D = -4 = \text{disc } \mathbb{Z}[i]$, then the principal form is $x^2 + y^2$. The field $\mathbb{Q}(i)$ has class number 1, hence trivial Hilbert class field, so we see that an odd prime p is represented by the form $x^2 + y^2$ if and only if it splits $\mathbb{Q}(i)$, if and only if $x^2 = -1$ has a solution modulo p , if and only if $p \equiv 1 \pmod{4}$.

Similarly the field $\mathbb{Q}(\sqrt{-2})$ ($\text{disc } \mathcal{O}_K = -8$) has class number 1, and p is represented by $x^2 + 2y^2$ if and only if the equation $x^2 = -2$ has a solution modulo p , if and only if $p \equiv 1, 3 \pmod{8}$.

The field $\mathbb{Q}(\sqrt{-5})$ ($\text{disc } \mathcal{O}_K = -20$) has class number 2, and its Hilbert class field is $\mathbb{Q}(\sqrt{-5}, \sqrt{5}) = \mathbb{Q}(\sqrt{-5}, i)$. A prime number $p \nmid 20$ is represented by $x^2 + 5y^2$ if and only if p splits in $\mathbb{Q}(\sqrt{-5}, i)$, if and only if the equations $x^2 = -1$ and $y^2 = 5$ have solutions in $\mathbb{F}_p$, if and only if $p \equiv 1, 9 \pmod{20}$. The other reduced form is $2x^2 + 2xy + 3y^2$, and a prime number $p \nmid 20$ is represented by this form if and only if p splits in $\mathbb{Q}(\sqrt{-5})$ but does not split completely in the Hilbert class field $\mathbb{Q}(\sqrt{-5}, i)$, if and only if $p \equiv 3, 7 \pmod{20}$.